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4.6 Excellent

Animal wrongs

Efforts to reduce animal testing mean the case for animal research must be restated

ALL of a sudden, governments have begun to talk about something that once seemed impossible: the end of animal testing. Last year, the US and the UK both unveiled road maps intended to help move us towards this goal. A few months ago, the European Commission did likewise. The coming of age of technological alternatives to some kinds of animal research (see page 32) is to be celebrated. A few of these alternatives are actually more revealing than the animal experiments they replace. But if nations overstate what is possible, they risk reigniting a heated and painful row.

We see this overstatement not only in news headlines, but even in the titles of policy documents themselves. Some of these talk of “phasing out” animal testing,

when the truth is that animal testing cannot and will not be ended entirely.

There will always be a need for certain kinds of animal research to ask fundamental biological questions and ensure drug safety. This means that, while pursuing the reduction in animal tests as

“Some scientists had their property set on fire. Molotov cocktails were used”

far as possible, we simultaneously need to be restating the case for why animal testing is so important.

To see what happens if we don’t, look back to the late 20th and early 21st centuries, when extreme animal rights activists violently harassed and

intimated scientists. Some researchers had their property set on fire. Molotov cocktails were used. National newspapers reported on calls to murder those carrying out animal experiments.

If we send out the message that animal testing can be phased out, we risk painting the scientists who continue it as wilfully causing harm to animals. Such a misunderstanding could take us back to those dark and violent times, while prematurely abandoning the kinds of animal tests for which there are no reliable alternatives would risk our health and safety.

So let’s be clear-eyed. The push to reduce animal testing where safe and possible is truly welcome. But the end of animal testing as a whole remains impossible. ■

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CONTACT US

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General & media enquiries

UK 9 Derry Street, London, W8 5HY
Australia 58 Gipps Street, Collingwood, Victoria 3066
US 51 Astor Place, 4th Floor, New York, NY 10003

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The New Scientist view

AI is both the best and worst thing to ever happen to mathematics

Mathematicians now have a formidable tool for breaking open challenging problems, but working out how best to use it isn't going to be easy



OUR best equations for describing the way fluids move were first written down about 200 years ago. Ever since, we have been unable to answer a simple question about them: do they sometimes blow up? This explosive phrase has a precise mathematical meaning describing a situation where, apparently out of nowhere, part of the fluid starts to move infinitely fast. It would be like an infinite whirlpool forming in your bathtub in response to a flick of your toes.

Understanding the Navier-Stokes equations – named after the pair who discovered them – has been a long-running quest for mathematicians. So much so that the blow-up question sits on the list of prestigious Millennium Prize Problems, each of which comes with a \$1 million reward.

This week, OpenAI stunned the world of mathematics with its announcement that its AI agents have finally found the answer: the equations do sometimes blow up (see story on the next page). The implication isn't that fluids will start misbehaving, but instead that the equations aren't perfect. This is good news for your bathtub. Mathematics, however, may never be the same again.

The reason for this isn't the result itself. The Navier-Stokes equations have already proven themselves to be incredibly useful many times over, in applications ranging from making better aeroplane wings to modelling the flow of fluids through artificial hearts. Mathematicians had a suspicion that blow-ups were possible. Now they know for sure.

That isn't to underplay the discovery – it is one of the biggest mathematical breakthroughs in decades. But the way it has happened, and the resulting

"It isn't just the Navier-Stokes equations that AI has blown up, but mathematics itself"

fallout, is what will have the most dramatic effect on mathematics for years to come.

OpenAI began working on the problem after it got wind that a pair of mathematicians had made some progress on it, which hadn't yet been published. Researchers at the tech company decided to set thousands of AI agents on solving the problem. A few days later, they were done.

Over the past few months,

It might have been quicker to just ask a chatbot

AI has made breakthrough after breakthrough in mathematics, with this most recent development being by far the most impressive. It wasn't that long ago that mathematics was a bit of an embarrassment for AI because it was so bad at it. Now it is shockingly good.

How many other long-standing mathematical conundrums will fall to AI is uncertain, but it will surely be many. OpenAI has already said it has made "substantial progress" on another of the Millennium problems.

But AI proofs aren't a straight replacement for human ones. Mathematicians want to know the why; AIs not so much.

The great mathematician Paul Erdős believed that mathematical truth exists independently of humanity in a sort of celestial book, and it is the job of mathematicians to discover what is in it. Sometimes, a mathematician would discover a logical argument that was so insightful and so elegant that it would bring a fresh understanding or a completely new way of thinking. Erdős would declare that a proof like this was "straight from The Book". It is fair to say that OpenAI's effort wouldn't make the cut.

That is because AIs tend to produce convoluted, hard-to-follow arguments. For the sake of proving a theorem, this doesn't really matter, thanks to the magic of a process called formalisation, where proofs can be turned into code that can be rigorously checked by computers. But for the sake of gaining insight, it is lacking.

So, where does this leave mathematics? Clearly, the AI

mathematical revolution has arrived. It is a new era where we will have many more answers than explanations, and unpicking it all won't be easy. It isn't just the Navier-Stokes equations that AI has blown up, but mathematics itself.

Vive la révolution

These are unprecedented times, and yet mathematics has gone through upheaval before. When René Descartes made the link between geometry and algebra in the 17th century, showing that shapes could be written as equations, he completely transformed the toolkit available to mathematicians.

Geometry, however, had been the darling of mathematics since the ancient Greeks. It was trusted and considered pure. Mathematicians worried that the job would make them drones who simply manipulated symbols and came up with no new insights. Philosopher Thomas Hobbes described the approach as a "scab of symbols".

But 300 years later, it is indisputable that making this link between algebra and geometry was a good thing. Without it, there would be no calculus or relativity – in fact, modern science would be completely unrecognisable. Perhaps AI will have a similar effect on the field.

Mathematicians now have an imperfect truth machine that spits out mathematical answers. There are big challenges ahead regarding how to use it, who gets access and what role AI companies should play, but it is a tool that has the potential to allow mathematicians to make progress much faster than before. AI may end up being the best and worst thing to have ever happened to mathematics. ■

Biggest breakthrough in decades

AI has stunned the world by solving one of the most prestigious problems in mathematics.

Matthew Sparkes and **Karmela Padavic-Callaghan** explain what it all means

OPENAI'S AI agents have found a solution to one of the toughest and most enduring puzzles in mathematics. The astonishing announcement came after an unusual flurry of activity around the problem, along with a dispute around how the work came about.

The Navier-Stokes equations describe how fluids move in space, such as air over an aircraft wing or water out of a tap. The equations remain mysterious in many ways long after being written down. "The equations themselves are so simple, but we understand so little," says John Evans at the University of Colorado Boulder.

To encourage mathematicians to develop a better understanding, the Clay Mathematics Institute chose resolving a particular question surrounding them as one of its seven influential Millennium Prize Problems. That question is to determine if the equations always work or if there are situations where they stop making sense and start spouting nonsense, what mathematicians call blow-ups.

OpenAI approached the problem by setting 1000 AI agents on creating blow-ups for Euler equations, which are a stepping stone to the full Navier-Stokes equations. It took the agents 50 hours to find and prove the existence of blow-ups in this simpler case. OpenAI researchers then set 10,000 agents the task of extending the blow-ups to apply to the full Navier-Stokes equations, which took just 11 hours to finish. Within 88 hours of the first agents kicking into action, a solution had been found.

The company said during a press conference that if a customer wanted to run the same problem, it would cost around \$15 million. It didn't name

the AI model used, but said it was "significantly more capable" than even its latest GPT-6 Astra model.

"This problem has remained unsolved for 200 years because the Navier-Stokes equations are just so enormously complex, and the pen-and-paper calculations you need to do in order to solve this problem are just mind-bogglingly intricate," says Venkat Chandrasekaran at OpenAI, who was also at the press conference.

The result is the latest in a string of shocking mathematical discoveries led by AI in recent months. In May, an OpenAI model cracked a decades-old conjecture by Paul Erdős, causing a stir in mathematical circles. Later, the Claude Fable 5 AI found a counterexample to the Jacobian conjecture, which had stood for nearly a century. Earlier this month, an AI model formalised (the process of turning proofs into code that can be checked by

computers) Fermat's last theorem in just 11 days, a mathematical task that was expected to take a team of humans around five years. Even amateurs have made breakthroughs, including one of the authors of this story (see page 17).

Sébastien Bubeck at OpenAI says the latest breakthrough is the "spectacular combination of the arc we have seen over

the last 12 months".

However, the mathematics the AI agents relied upon has been in the works for far longer than that.

In 2014, Thomas Hou at the California Institute of Technology and his colleagues found a way to create a blow-up for Euler equations, but their approach applied only when the fluid in question was walled in, and not more generally. Other teams tried and failed to extend it. Researchers started to suspect that the Navier-Stokes equations really can blow up, but no one had definitively shown how to construct a specific fluid flow that does so.

A breakthrough came in 2021, when Luis Martínez-Zoroa at CUNEF University in Spain and Diego Córdoba at the Spanish National Research Council developed a new method for creating blow-ups that worked for both Euler and Navier-Stokes equations. Their key insight was to divide the fluid into thin layers and perform calculations for each. Adding up the layers resulted in a cascade of effects that can ultimately cause a blow-up.

"This is really the showstopper," says Steven Brunton at the University of Washington. "AI [use] was absolutely pivotal, and the scale [of it] is interesting and useful, but the solution approach was developed by brilliant humans."

The one problem was that Martínez-Zoroa and Córdoba's work relied on an unrealistic force being used, which means it wouldn't solve the Millennium problem.

Then, last week, Tristan Buckmaster at New York University and Levent Alpöge at AI company Anthropic published a breakthrough removing that requirement for Euler equations and several related situations, paving a clear path to Navier-

88 hours

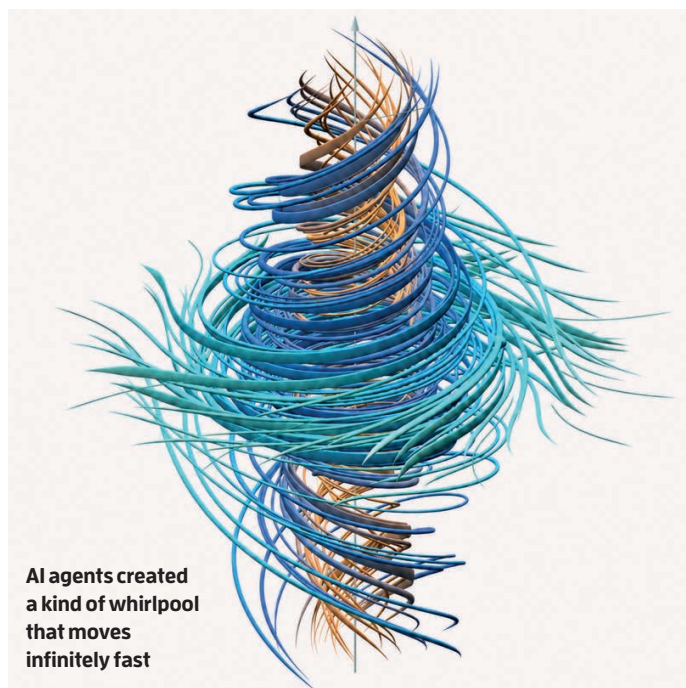
Time it took OpenAI's AI agents to solve the Navier-Stokes problem

10,000

The number of AI agents that contributed to the result

\$15m

The estimated spend on AI the work would have cost a customer



AI agents created a kind of whirlpool that moves infinitely fast

OPENAI

A controversial race to the finish

Tainting the excitement of OpenAI's mathematical result has been questions around academic integrity and exactly how the work came about.

Shortly before OpenAI released its news, a pair of researchers, Tristan Buckmaster at New York University and Levent Alpöge at AI company Anthropic, rushed out their solution to a close cousin of Navier-Stokes, the Euler equations, which opened the door to a full solution (see main story).

Buckmaster wrote in a document posted to his website that, as a result of rumours that

OpenAI was making progress on the problem, he emailed a prominent mathematician at the firm to clarify matters. He says that in a subsequent meeting with OpenAI staff, he was given no hard details about how OpenAI achieved the result, but he took what little he was told as a "red flag", as it appeared to describe the same technique and path that he and Alpöge had followed to achieve their interim findings.

According to Buckmaster, OpenAI then admitted it hadn't started its own AI search for a solution until after it had heard

of his and Alpöge's work, and didn't respond to questions about whether its AI model had been trained or fine-tuned on that existing work – which was being stored in OpenAI's Codex model, where the company would theoretically have access to it. The pair had used Codex as a customer, not a research partner, so expected that their work should remain secure and private.

In a press conference announcing its discovery, OpenAI categorically denied using Buckmaster and Alpöge's proof or prompts in its work. It also

said that its model's proof of Euler is different from that put forward by Buckmaster and Alpöge. It also said that no humans accessed Codex to see work in progress within.

Mark Chen at OpenAI, when asked if that also applied to any of the thousands of AI agents that had worked on the problem, said "that's also our understanding".

The company also wrote on X: "While unlikely, we cannot rule out that de-identified data derived from their usage of our products helped improve our models."

MS

Stokes. A few hours afterwards, OpenAI published its full result. (For the wider story surrounding this, see "A controversial race to the finish", above).

OpenAI's agents used an approach like Martínez-Zorúa and Córdoba, working iteratively to build a blow-up. However, the AI's trick was ultimately distinct, not just a more powerful repetition, says Evans.

The specific blow-up was a spinning whirlpool that becomes longer, thinner and more intense

"I have colleagues who will think it's the end of the world – I'm in the 'a bit terrified' camp"

over time, before its swirling ultimately becomes infinite (pictured on page 5).

The new proof showed that this happens in finite time and using only a finite amount of force on the fluid. "We were very surprised by the developments: both Diego and I were confident that the resolution was a matter of time,

but we weren't expecting it to be so soon," says Martínez-Zorúa.

Evans says that it is exciting that, in the work from OpenAI, most of the key ideas the AI agents used are relatively simple ones that an undergraduate could have learned in class. "Even though it's 166 pages of complicated math, you can actually draw it down to a couple of simpler physical mechanisms," he says. In his view, AI agents could help make better models of fluid flow in the future.

Brunton says that the Navier-Stokes equations will continue to be useful. "Every equation is an approximation. Navier-Stokes is a great approximation for a very big range of fluids, but at the point it starts to blow up to infinity, it's not a good model of the world any more," he says.

The Navier-Stokes equations model fluids as continuous entities rather than being made of individual atoms, so at very small scales, a different mathematical model is necessary. However, for engineers, that level of precision won't be needed, says Brunton.

"Nothing changes for us," he says. For mathematicians, however, the changes could be massive.

Helen Wilson at University College London says that academics are still processing what is happening. "With any new development, I have colleagues who I can guarantee will think it's the end of the world, and colleagues who I can guarantee will think it's the best thing since sliced bread, and this one's no different," says Wilson. "I'm in the 'a bit terrified' camp."

Take the question of how to train new mathematicians. Currently, PhD students are often given a relatively low-level problem to chew over for a few years and hone their skills on, but that model falls apart with AI. "If the world's moving so fast, there won't be anything that they can do slowly, because it'll all get done before they finish," says Wilson.

And in a world where we fail to

The Navier-Stokes equations have helped build better aeroplanes

train fresh mathematicians, we soon run out of them. Within decades, the field becomes rudderless, and there are no humans left with the skills and experience needed to interpret what AI creates. It is said that no human has understood all of mathematics since Henri



Too much of a good thing

The way AI companies are approaching mathematics is harming the field, says **Terence Tao**

Poincaré, who died in 1912, but there may soon be parts of maths that no human understands.

"If there are no mathematicians, then those results don't go anywhere. It's just an industry of machines producing papers that nobody reads," says Wilson.

Some mathematicians fear for their careers in a world where computers can solve problems that they can't. But that is a reason to be optimistic, too. AI seems able to use existing knowledge at rapid speed to crack some of the meatiest mathematical conundrums. Perhaps mathematicians will prosper in the future as the controllers of such powerful tools, pushing the field further than they ever thought possible.

"We have so many questions that we don't know the answer to, and where it was extremely difficult to make progress – like really a struggle of a lifetime," said Bubeck, speaking to *New Scientist* before the Navier-Stokes announcement. "Now we're gonna be able [to answer them]." ■

TERENCE TAO is a man who understands mathematics. He has a Fields medal – often called the Nobel prize of mathematics – to name just one of his many accolades. So, when he says that OpenAI's recent blockbuster mathematical breakthrough is indicative of how big tech companies are negatively affecting the field, people are likely to listen.

Tao says the way in which discoveries are being made and "dumped" on the mathematical community may actually harm mathematics.

There has always been competition among mathematicians to be first, he says, which was fine because mathematics was so difficult that it created a natural brake to stop things getting out of hand. "Now, there's no speed limit, and suddenly things are breaking down," he says.

Tao fears that new findings will keep landing at an increasing pace and that there will be no time to absorb the results, fully understand them, put them into textbooks and teach them to students. This means the staggering ability of AI to bring us new mathematics could actually harm the field, instead of boosting it.

Mathematics needs to overcome five hurdles in order to become useful, he says. First, it needs to be created, then checked, explained, accepted and finally digested into the corpus of work that is taught to the next generation. He says AI firms are only concerned with the first two.

"These companies are dumping carcasses of raw meat onto our communal village table and saying: 'Here you go, I solved your food problem.'



DAVID ESQUIVEL/UCLA

Profile

Terence Tao is based at the University of California, Los Angeles, and is often described as the world's leading mathematician

And then they just leave."

As an example, he points to the situation surrounding Tristan Buckmaster at New York University and Levent Alpöge at AI company Anthropic (see "A controversial race to the finish", left). They had been "working slowly on some very nice results" around the Navier-Stokes equations, says Tao, and had developed a solution to the Euler equations, a close cousin of Navier-Stokes that would act as a stepping stone to solving them.

But they say they were forced to suddenly rush out papers – which mathematicians told *New Scientist* lacked the clarity they would normally expect – once they found out that OpenAI had overtaken them. Ironically, OpenAI only began working on the Navier-Stokes problem once it heard rumours of Buckmaster and Alpöge's work.

Tao says that what has developed is a race that benefits nobody (except perhaps the marketing departments of AI companies), erodes clarity and

prioritises who is first over who is accurate, clear and concise.

"This is not how science is supposed to work," says Tao. "They're focusing on what traditionally has been the most prestigious part of mathematics, but then there's sort of the lower-prestige work of explaining it, communicating it, teaching it, refereeing it for journals – all that work is left to us to clean up, and it's demoralising."

Increased competition from AI will also jeopardise the open, sharing culture that exists in mathematics and push research underground, he says, for fear of others swooping in on their ideas and completing them before they can publish.

"It used to be that, if you announced you were working on a problem, there were maybe only a dozen people in the world who could plausibly outcompete you, and you knew all of them, you were on a first-name basis. There was no incentive to screw over

"These companies are dumping carcasses of raw meat onto our communal village table"

your colleagues because they would screw you back," says Tao. "Knowing that someone's working on a problem is now valuable information"

Despite his pessimism, Tao believes that mathematicians will eventually do the hard work of understanding, categorising and teaching AI company output.

"It will eventually end up in textbooks, but it's just a completely sordid way to get there," he says. ■

MS



ANDRZEJ PIŁOWALAMY

Pink noise helps clear brain waste

The brain flushes out waste products while we sleep, and now it seems that certain sounds might enhance the process, finds **Carissa Wong**

SLEEPING while listening to bursts of pink noise – background sounds akin to gentle radio static – seems to boost the flow of fluid through the brain, and so might enhance the clearance of waste products that are linked to conditions like Alzheimer’s disease. The pink noise seems to bolster the slow brainwaves that enhance the pumping of blood vessels, which drives the brain’s waste-disposal system.

We already knew that listening to bursts of pink noise during sleep can strengthen slow brainwaves, a kind of electrical activity that occurs during certain phases of non-REM sleep, known as N3 (deep sleep) and N2 (the lighter stage before it). This effect occurs when the bursts coincide with the peaks of slow brainwaves.

“They really don’t sound like much; they’re just little staticky beeps,” says Joshua Levitt at Boston University in Massachusetts.

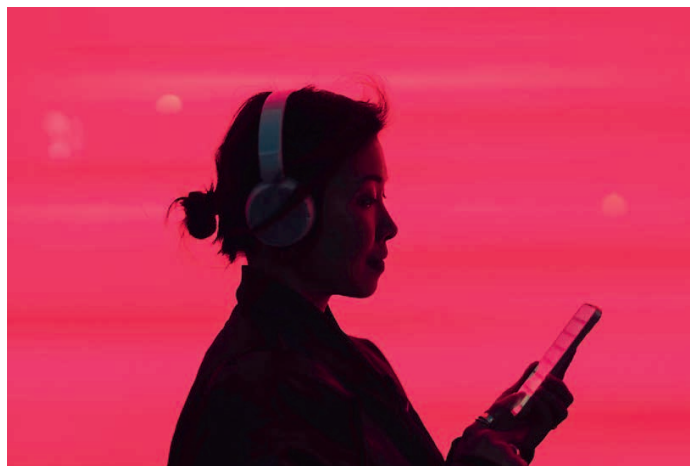
Prior research has also shown that boosting slow brainwaves – using drugs, for instance – can enhance the flow of the brain’s cerebrospinal fluid (CSF), which bathes the brain and carries

away waste products.

But it was unknown whether listening to pink noise has the same effect. That’s because measuring the brain’s CSF flow involves imaging it with MRI scans, but MRI interferes with EEG, the technique used to measure waves of electrical activity in the brain. One outcome of this interference is that slow waves recorded via EEG during MRI scans can’t be detected fast enough to align bursts of pink noise with their peaks.

“You have to do a lot of extensive processing of the EEG signals to detect slow brainwaves, so it’s hard to do that fast enough in real time,” says Laura Lewis at the Massachusetts Institute of Technology.

To overcome this problem, Levitt, Lewis and their colleagues used EEG data collected during MRI scans from prior studies to train an AI model to rapidly predict when the brain’s slow waves will peak. “[The AI] says, ‘In about 70 milliseconds, a slow wave peak is coming,’ so now we should schedule our sound stimulation to arrive in accordance with that,” says Levitt.



OSCAR WONG/GETTY IMAGES

New
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Hear Kimberley Wilson discuss how to build lifelong brain health on 10 October
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“This, methodologically, really moves the field forward,” says Sephira Ryman at the University of New Mexico.

The researchers then recruited 27 healthy adults, aged 29 on average, to wear EEG electrodes on their scalp while taking an afternoon nap in an MRI scanner. Of these, 14 managed to fall asleep and enter N2 sleep.

At the peak of half of each participant’s slow brainwaves, the researchers used the AI model to play 50-millisecond bursts of pink noise. During the remaining slow brainwaves, they played no noise as a control.

By analysing the MRI and EEG recordings, they found that bursts of pink noise strengthened the slow waves they coincided with

Zapping the face may improve brain signalling

Beyond finding ways of enhancing the brain’s waste-disposal system, (see main story), research has also found that electrically stimulating a nerve in the face shows promise for boosting memory. Bursts of stimulation appear to improve signalling between cells in the hippocampus, a brain region crucial for memory.

Myles Mc Laughlin at the Leuven Brain Institute in Belgium and his colleagues previously showed that a non-invasive electrical-stimulation technique sparks the trigeminal

nerve, a nerve in the head that carries sensory information between parts of the face and the brain. This then activated a pathway leading to the hippocampus.

Now, the team has tested if this affects memory. They first electrically stimulated the trigeminal nerve of anaesthetised rats, via electrodes under the skin. They found that 1-second bursts of 200-hertz stimulation, separated by 29 seconds of rest, increased signalling between hippocampal neurons by about 20 per cent. This

continued for at least 90 minutes after the stimulation ended.

The researchers think intermittent stimulation activates the locus coeruleus, the brain’s primary producer of noradrenaline, or norepinephrine – a chemical messenger that helps transmit nerve signals between neurons, and may also enhance memory.

The team then tested the effects of this on memory. Thirty-six healthy adults completed one session of bursts of stimulation, delivered via skin electrodes, and

one sham session, at least two days apart. In the sessions, they had to remember the names and jobs that went with 12 faces, which they were tested on 2 minutes later. They then received bursts of stimulation for 20 minutes, before being tested on their memory again.

The stimulation improved job recall, but only at the 20-minute mark (medRxiv, doi.org/rn2t). The improvement corresponds to “roughly one extra association being retained”, says Mc Laughlin. Rachel Melegrito

Ancient humans

Denisovan tools show how the mysterious humans lived

Michael Marshall

Pink noise plays lower frequencies, so isn't as high-pitched as white noise

and briefly boosted the flow of CSF into the brain, compared with no sonic stimulation. This suggests the pink noise also enhanced the flow of CSF through, and out of, the brain. "We know that the flow in is usually balanced with the flow that comes out," says Lewis.

Cleaning out proteins

The increased CSF flow seemed to be driven by enhanced pumping of the brain's blood vessels, which is known to push fluid through the brain's waste-disposal system, called the glymphatic system (*Science Translational Medicine*, doi.org/hch554).

Off the back of these results, the team is exploring whether the approach also works in older adults and whether it can boost the clearance of proteins, such as beta-amyloid, that are linked to conditions like Alzheimer's disease. If the results are positive, the researchers hope to test whether the technique can slow cognitive decline in people during normal ageing, mild cognitive impairment – a condition that often precedes dementia – and the early stages of Alzheimer's disease.

There is reason to think this could work. Exposing people to sounds and flickering lights – while they are awake – has previously shown promise at slowing cognitive decline in people with Alzheimer's, potentially by boosting the glymphatic system.

But one benefit of the new approach is that it could eventually be delivered via a portable device while people are sleeping, which could be less disruptive, says Ryman. ■

A GROUP of enigmatic ancient humans called the Denisovans thrived in what is now south-west China for more than 100,000 years. It is the first evidence of Denisovans in this region, and one of the only known sites with artefacts that can tell us about how they lived.

The Denisovans were hominins that lived in East Asia within the past few hundred thousand years. However, it is impossible to study the archaeological context of almost all the remains currently discovered. Because of this, we have little information about the tools and other artefacts they made.

The new finds come from Bianfu cave in Yunnan province. Heavy quarrying in the area has destroyed much of the cave, but in 2019 workers spotted animal fossils and alerted the authorities. This led Hao Li, an archaeologist at the Institute of Tibetan Plateau Research in Beijing, and his colleagues to conduct an excavation that summer.

They dug three trenches to a depth of 6.6 metres, revealing

18 layers of sediment – of which 10 contained artefacts made by hominins. One of these layers yielded four intact hominin teeth.

The excavation also unearthed over 64,000 fragments of bone, which a team led by Qiaomei Fu at the Institute of Vertebrate Paleontology and Paleoanthropology in Beijing inspected by hand to look for potential hominin remains. The researchers identified 22 candidates, which they subjected to molecular analysis. Three proved to be from

"The Denisovans clearly had a very skilful hunting strategy to be able to bring down deer"

hominins: two skull fragments and a piece of radius, an arm bone (*Nature*, doi.org/rnwk).

They tried extracting DNA from the three bones, but to no avail. Fu says the warm climate in the region and the age of the fossils were "unsuitable for DNA preservation". However, the team did manage to obtain proteins from the three bones and two of the teeth. All five held protein variants that are unique to Denisovans, and none had any protein variants specific to *Homo sapiens* or Neanderthals. Fu and her colleagues therefore determined that all the remains came from Denisovans.

The bones and teeth appear to be different ages, so Li's team concluded that they come from at least four individuals. The two studied teeth had telltale protein variants showing they belonged to males.

The researchers dated the bones themselves and the surrounding sediments using multiple methods. They determined that the bones are

between 134,000 and 167,000 years old. However, the artefacts cover a longer time frame: 70,000 to 190,000 years ago.

"We think this is one hominin species who occupied the site for a long time," says Li. The excavators found a total of 1366 stone artefacts. These included 135 flakes and the cores from which they had been removed, along with simple tools such as scrapers. In general, the artefacts were relatively simple, compared with those made by Neanderthals living elsewhere in Eurasia at the same time.

However, the Denisovans also seem to have been successful hunters. The team found 1654 identifiable animal bones, including large numbers of deer and bovids (hoofed animals that include cows and gazelles), plus rhinoceroses and hyenas. Many of the bones had telltale marks suggesting they had been butchered by hominins. Deer and bovids were the main prey (*Nature*, doi.org/hch7m3).

Li says the Denisovans clearly had a "very skilful hunting strategy" to be able to bring down such prey. He argues they probably relied less on stone tools than other hominin groups, explaining why the stone artefacts they found were relatively crude. The team discovered seven bone tools, and Li suspects the Denisovans used these a lot.

This may reflect the comfortable environment in which these Denisovans lived. "It's a different ecosystem from where we had them previously," says Bence Viola at the University of Toronto in Canada. Pollen evidence suggests the area was a mix of conifer forest and steppe, with a temperate climate and rich wildlife. ■

The tools were simple compared with those made by Neanderthals



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Reproductive health

Paracetamol use in early pregnancy linked to smaller ovaries and testes

Michael Le Page

GIRLS whose mothers took paracetamol (acetaminophen) during pregnancy had smaller ovaries at three months old, on average, than those whose mothers didn't take the drug, according to a study of about 300 infant girls in Denmark. Infant boys whose mothers took paracetamol had smaller testes on average too.

That doesn't prove paracetamol caused the reduction in the average size of ovaries and testes, though – this association could be a result of the conditions that led to the women taking paracetamol during pregnancy, or even due to chance. Nor is it clear whether having smaller ovaries or testes as an infant affects fertility in adulthood. However, the findings are in line with separate studies done in rodents, in isolated human ovaries and in human embryonic cells.

"We still recommend that pregnant women should use

paracetamol with high fever or strong pain," says Margit Bistrup Fischer at Copenhagen University Hospital – Rigshospitalet. Not doing so can be risky, she says.

But most of the women in the study used paracetamol for pain, rather than fevers, and could have managed their discomfort in other ways, such as heat treatment or physiotherapy, says Fischer.

"We should have a dialogue about when paracetamol is actually necessary," she says. "It is still a drug and it does have side effects."

The issue is that there are almost no safe drug-based alternatives to paracetamol for pregnant women. Most other painkillers aren't recommended because they can cause congenital conditions.

Fischer's team recruited 685 pregnant women during their first trimester, who were asked about their use of medicines every two weeks. They had a fetal ultrasound

during their third trimester, and 302 daughters were examined around three months after birth.

Those exposed to paracetamol as a fetus had, on average, 40 per cent smaller ovaries by volume, 13 per cent smaller uteruses by volume and 23 per cent fewer ovarian follicles – the sacs

"Girls exposed to paracetamol as a fetus had, on average, 13 per cent smaller uteruses"

that nurture immature eggs – compared with those who weren't exposed. Higher doses were linked with a bigger effect, says Fischer, but no women took more than the recommended daily maximum of 4000 milligrams (*Human Reproduction Open*, doi.org/rn9f).

The team also examined the boys who were born to the 685 women. "We do see a difference,

on average, in the boys as well," says Fischer. "[They] have a reduced testicular volume."

Even so, other experts say these results are not definitive. "It is still too early to sound the alarm, and to remove this tool from the therapeutic arsenal for treating pain and fever," says Séverine Mazaud-Guittot at the National Institute of Health and Medical Research in France.

But this study, along with one from last year showing that paracetamol affects the proliferation of human embryonic cells, suggests that the precautionary principle should be applied, especially during the first trimester, says Mazaud-Guittot. "Raising awareness of alternative, non-pharmacological methods should help to encourage a reduction in the use of medication for the treatment of various mild pains," she says. ■

Palaeontology

Earliest alligator thrived after dinosaur extinction

A 2-METRE-LONG reptile that flourished in North American waterways shortly after an asteroid wiped out most dinosaurs 66 million years ago has been identified as the oldest known representative of the alligator group.

At Corral Bluffs in central Colorado, researchers uncovered bones from about 15 individuals of the previously unknown crocodilian species, which probably ambushed-hunted near freshwater channels.

The newly named *Mandrasuchus milleri* looked like a smaller version of a modern alligator, with pointed front teeth for seizing prey and rounded, molar-like back teeth



ANDREY ATUCHIN

that could crush shellfish and other hard-bodied prey.

These characteristics may help explain how it survived in a dying world. Its success may then have paved the way for the widespread expansion of alligators, says Sadie

Sherman at the Denver Museum of Nature & Science in Colorado.

"We have a bunch of these specimens, so they were quite abundant," she says. "And so, to have a large-bodied apex predator not just living, but actually thriving,

Mandrasuchus milleri is the oldest known alligator species

so soon after such an apocalyptic event is really telling."

Analyses of fossil pollen and radiometrically dated volcanic ash revealed that the oldest skulls date to 350,000 years after the asteroid strike, making *M. milleri* the oldest known member of the alligatorine subfamily (*Journal of Vertebrate Paleontology*, doi.org/rn89).

"*Mandrasuchus milleri* provides a rare insight into how alligatorids diversified almost immediately after the end-Cretaceous extinction," says Adam Cossette at New York Institute of Technology in Long Island. "The fossils are compelling." ■

Christa Lesté-Lasserre

Feedbacks amplify global warming

Climate models haven't fully accounted for natural feedback loops such as the release of greenhouse gases from wetlands and permafrost, finds **Alec Luhn**

FEEDBACK loops that cause ecosystems to emit more carbon will make the climate crisis more severe than expected, amplifying global warming by 20 to 30 per cent this century.

It isn't just fossil fuel emissions that raise temperatures. Warming itself creates more warming. For instance, a hotter atmosphere absorbs more water vapour, which traps more heat in the atmosphere, which then holds more water vapour.

Climate models already incorporate most global feedback loops like this one, and they have started to include feedback mechanisms that are disrupting carbon storage in particular ecosystems. But no single model includes all ecosystem feedbacks. Now, a study has shown that these natural feedbacks will heat the planet by an extra 0.2°C to 0.4°C by 2100, on top of at least 1°C to 2°C of global warming already expected between now and then.

"That suggests we're going to go further over 1.5°C global warming than we thought, with all the extra risks that brings," says Tim Lenton at the University of Exeter in the UK, who wasn't involved in the new research.



SHUTTERSTOCK/MIKE WORKMAN



Hear more about nature and the climate crisis from three of the world's leading scientists on 11 October
[newscientist.com/nsimag](https://www.newscientist.com/nsimag)

Christina Schädel at Woodwell Climate Research Center in Massachusetts and her colleagues estimated the emissions from four major natural feedbacks. The first is mainly in the Arctic, in permafrost that once remained frozen all year. As this ground thaws, microbes begin to decompose plant and animal remains that have been

frozen for millennia, emitting carbon dioxide and methane.

A second feedback involves wetlands, which also release more methane as temperatures rise. This is because warmer water can hold less gas, and because microbes in the water decompose organic matter at a higher rate. The same processes lead to

Thawing permafrost in the Arctic is one of four major natural feedbacks

additional methane emissions from ponds and lakes, a third feedback. The fourth feedback kicks in as the air gets hotter, winters get shorter and soils get drier. All of this increases the frequency and intensity of wildfires, which results in greater CO₂ and methane emissions.

Accelerated change

"We need to fully account for all sources of emissions that contribute to warming, otherwise we plan for the wrong future," says Schädel. "These emissions are left out of carbon budgets, and we adapt to the wrong targets."

Much of this extra warming is already unavoidable, the modelling suggests. In a climate scenario in which human emissions decline after 2050, the natural feedbacks add an average of 0.4°C of warming by 2100.

But even in a scenario in which emissions start declining today, the feedbacks add 0.2°C of extra warming this century, prolonging global warming (*Environmental Research Letters*, doi.org/hch6xh).

"This additional warming keeps rising through 2100... because there is a lag between when emissions occur and when the temperature impacts occur," says Schädel.

Rather than instilling hopelessness, this should inspire governments to redouble their efforts to cut emissions, she adds. Currently, fossil fuel emissions are still increasing each year.

The new findings mean governments need to invest even more in measures like drought-resistant crops and early warning systems for climate tipping points, says Lenton. ■

World's biggest carbon sink may start emitting CO₂

Another source of carbon dioxide that could cause extra warming (see main story) is currently the world's largest carbon sink. The Southern Ocean will start releasing CO₂ if the world reaches net-zero emissions and atmospheric CO₂ levels fall, a modelling study warns.

Strong winds around certain southern latitudes encourage the ocean to absorb CO₂. Cold Antarctic air cools the water and causes it to sink, storing that carbon away in

the deep ocean. But if levels of CO₂ in the atmosphere begin falling, there will be a higher concentration of CO₂ in the surface water than in the air, and the Southern Ocean will start releasing rather than absorbing CO₂.

Yechul Shin at Seoul National University in South Korea and his colleagues modelled a scenario in which the world reaches net zero in 2125 and the concentration of atmospheric CO₂ starts to gradually

decline as Earth's natural carbon sinks remove some of the gas from the atmosphere. They also modelled one in which humanity drastically decreases atmospheric CO₂ after net zero – again by 2125 – via carbon dioxide removal (*Science Advances*, doi.org/hcjfxp).

In both scenarios, the Southern Ocean goes from absorbing about 3 grams of carbon per square metre per year now to emitting 8 to 9 g of carbon per m² by 2400. **AL**

Biology

Immune cells may kick-start birth

Experiments suggest transfer of mitochondria provides uterus with energy for contractions

Alice Klein

WHAT sets off labour when a baby is ready to be delivered has long been a mystery, but now it seems that the immune system pulls the trigger. The uterus appears to start contracting after immune cells transfer energy to its muscle cells via tiny tubes.

“By understanding all the steps along the road, we’re getting closer to identifying therapeutics for preventing preterm birth,” says Stephen Lye at Mount Sinai Hospital in Toronto, Canada.

Lye and his colleagues previously found that, towards the end of pregnancy, the human uterus releases signalling molecules called cytokines and chemokines into the bloodstream. These usher immune cells called monocytes from the blood into the uterus. The monocytes accumulate in the muscle tissue of the uterus and turn into other immune cells called macrophages.

In the latest study, the team investigated the role of these macrophages by incubating them with muscle cells from uterus biopsies collected from pregnant women undergoing Caesarean sections at full term. “Remarkably, we found that the macrophages

“The influx of mitochondria also increased levels of an enzyme that breaks down progesterone”

made these little tube-like structures that attached to the muscle cells,” says Lye.

Even more remarkably, the team could see the macrophages sending mitochondria through these tiny tubes into the muscle cells. “In the images we took, you can see tons and tons of these mitochondria passing into the muscle cells,” says Lye.

Mitochondria are probably

transferred to muscle cells in the uterus to give it enough energy to contract and push the baby out, says Lye. In line with this, the team found that levels of the energy-providing molecule adenosine triphosphate increased in the uterine muscle cells after the macrophages connected their mitochondria-delivering tubes to them.

The influx of mitochondria also increased levels of an enzyme that breaks down progesterone. This probably signals for labour to begin, because we know a drop in this hormone turns on the genes required for labour (bioRxiv, doi.org/rn2q).

“In virtually all species, in order to go into labour, you have to get rid of progesterone,” says Lye. “Most species do this by stopping the production of progesterone, but in humans, progesterone levels in the circulation don’t fall

by the time of labour. So for decades, people have been saying: ‘Why is it that women actually go into labour?’” he says. The new results suggest that human labour is specifically activated by a drop of progesterone in the muscle cells of the uterus, not the whole body, he says.

The researchers hope to use their findings to develop more effective interventions for preventing preterm births, which occur before 37 weeks of gestation. Being born before this time, when organs haven’t fully developed, increases the likelihood of conditions such as cerebral palsy, and vision and hearing problems.

Lye and his colleagues are now conducting a study looking at various immune cells in the blood of those who present with possible signs of early labour, to see if they can identify who is really at risk of delivering. ■

Palaeontology

Exquisite bird feather preserved in dinosaur dropping

A BIRD that was eaten and partially excreted by a dinosaur 66 million years ago has left behind the best-preserved feathers ever recovered from dinosaur-age rock. This pushes back the evolution of feathers like the ones on modern birds by 10 million years.

In 2016, David DeMar Jr at the University of Washington in Seattle discovered a broken coprolite (a fossilised dropping) with a feather clearly visible along its fractured surface in north-eastern Montana’s Hell Creek Formation. “I couldn’t believe what I was seeing,” he says.

He and his team later analysed the coprolite, which may have come



O’CONNOR ET AL.

from a young *Tyrannosaurus rex*, using micro-CT scans and 3D surface scanning. Two leg-bone fragments, 1.8 and 1.2 millimetres long, resembled some previously found at Hell Creek that were attributed to hesperornithiformes.

These were flightless, short-winged diving birds, similar to modern loons or grebes, and were related to the ancestors of modern birds.

Further analyses showed that the 9.3-millimetre-long primary feather had a water-repellent

The finding pushes the evolution of feathers back by 10 million years

structure, similar to that of modern pelicans, says team member Jingmai O’Connor at the Field Museum in Chicago.

Both this and a second, smaller wing feather from the fossil also had lightweight, spongy cores surrounded by a harder outer layer. This makes them both light and stiff, which is typical of modern feathers (*Current Biology*, doi.org/rn2h).

“What is exciting is to push the occurrence of modern feathers into non-modern birds,” says O’Connor. Differences in plumage are emerging as a major contributing factor as to why some birds survived the end-Cretaceous mass extinction while others didn’t, she says. ■

Christa Lesté-Lasserre



Hear Steve Brusatte talk about how one lineage of dinosaurs survived to this day on 11 October
newscientist.com/nslmag

Space

Gamma ray burst may have been two supernovae in one

Matt von Hippel

ON 1 January 2022, the Neil Gehrels Swift Observatory that orbits Earth spotted a huge burst of gamma rays from a distant galaxy. After four years of analysis, a group of astronomers has proposed the burst was a collapsing star triggering the formation of both a pulsar and a black hole.

According to a team led by Remo Ruffini at the International Center for Relativistic Astrophysics Network in Italy, the event began with the collapse of the core of a massive older star, which had earlier been transformed via nuclear reactions into a mix of carbon and oxygen.

The researchers argue that when the core exploded, an accelerating cloud of electrons and positrons was released in a hypothetical event that they have called an “HB supernova” after the late physicist Homi Bhabha. This cloud then ran into a nearby white dwarf, causing it to collapse and become a neutron star. The white dwarf’s collapse triggered a second supernova blast 3.5 seconds after the first.

Then, this second blast hit a third nearby object, another neutron star. As a result, the neutron star released a further burst of energy, the brightest of the entire event, before collapsing to become a black hole.

Meanwhile, fast-moving material continued to fall onto the new neutron star, causing it to spin faster and faster. This caused it to become a pulsar, a star whose fast spin and magnetic field produce regular beams of light (arXiv, doi.org/rnxc). “It’s the most complex system ever observed in astrophysics,” says Ruffini.

But other experts aren’t so sure. Although it is a fascinating idea, “the observations don’t definitively support this novel model”, and could also well fit a story that ends with a single black hole and no neutron star, says Ryan Ridgen at the University of Canterbury in New Zealand. ■

Environment

Super El Niño set to cause ‘terribly dangerous’ drought in the Amazon

Madeleine Cuff



CARL DE SOUZA/REX VIA GETTY IMAGES

A SUPER-SIZED El Niño weather pattern in the Pacific Ocean is set to cause a devastating drought in the Amazon, fuelling the risks of huge wildfires in a region already feared to be approaching a climate tipping point.

El Niño is a natural climate phenomenon that occurs when sea-surface temperatures in the tropical Pacific rise above normal. El Niño tends to push up global temperatures and unleash extreme weather globally.

Scientists monitoring the current El Niño’s development say it is set to be the most intense in modern history by a large margin. Very strong El Niños like this tend to concentrate their heat close to the coast of South America, says Adam Scaife at the Met Office, the UK’s national weather service. That, in turn, leads to more intense impacts in the region, including drought in the Amazon, he says.

The most up-to-date forecasts from the Met Office show a “terribly dangerous [rainfall] anomaly” developing over the Amazon by January, says Scaife. Hotter air temperatures are

also expected in the region.

The Amazon rainforest plays a key role in removing carbon dioxide from the atmosphere and is a globally significant biodiversity hotspot. But severe drought can have dire consequences for wildlife and the planet. An extreme Amazon drought during the 2015-16 El Niño event fuelled huge wildfires, destroying around 2.5 billion trees and plants, and emitted 495 million tonnes of CO₂, according to work led by

“The rainfall anomaly this summer is already much stronger than all previous large El Niños”

researchers at the University of Oxford and the University of Lancaster, UK.

Scientists also warn that repeated and increasingly severe droughts mean the Amazon is approaching a climate tipping point at which the rainforest ecosystem collapses and is replaced by savannah.

Tim Li at the University of Hawaii thinks the Amazon could

The very strong El Niño is fuelling the risk of huge Amazonian wildfires

suffer the most extreme El Niño-induced drought on record.

Rainfall over the Amazon during June, July and August was well below average, despite the El Niño phase still being in its infancy. “The rainfall anomaly in this year, 2026, in the summertime, is already much stronger than all other events [previous large El Niños]. In the Amazon, you already see drought,” says Li.

Assuming this year follows the pattern of previous large El Niños, that rainfall pattern will only intensify over the coming months, he says.

Meanwhile, air temperatures in the Amazon region are already much higher than normal, which, coupled with drought, will create a vicious cycle, says Li. “Once you have much less rainfall, then there’s positive feedback: no moisture, no evaporation, so the temperature gets even warmer.”

David Galbraith at the University of Leeds in the UK says the developing El Niño is “really, really worrying” for the Amazon. If it turns out to be as intense as predicted, the resulting tree die-off, reduced tree growth and intense wildfires “could cause losses of large areas of forest”, he says.

Research published last week by Jos Barlow at the University of Lancaster, UK, and his colleagues warns that parts of the Amazon are experiencing a rapid increase in heat and water stress, with the most pristine areas of jungle among the worst affected.

“The rates of change that we are seeing in the most affected regions are very alarming,” said Barlow in a statement. ■

Archaeology

Earliest evidence of habitual drug use

Alice Klein



TEETH uncovered in Indonesia that are up to 25,000 years old show signs of their owner having sucked on whole betel nuts, probably to release pain-relieving compounds. The finding is the oldest archaeological evidence of habitual drug use in humans.

About 10 years ago, Adam Brumm at Griffith University in Brisbane, Australia, and his colleagues were excavating a cave on the island of Sulawesi when they found a fragment of human jaw bone with three teeth. Dating methods revealed it to be 16,000 to 25,000 years old. "What really intrigued me was the presence of these bizarre grooves in the teeth," says Brumm.

The team then excavated a different cave on the island and found more teeth, this time about 6000 to 8000 years old, with the same unusual marks.

The rounded grooves were consistent with repetitively moving a hard, round object inside the mouth, and probably sucking and gnawing on it. This must have been done regularly, because the teeth were so worn down in some places that their nerve centres were exposed, which would have caused severe pain, says Brumm.

The skull of a hunter-gatherer who sucked on betel nuts habitually

One possibility was that the damage was related to an addiction to betel nuts, since they are a popular stimulant in modern Indonesia. However, the current practice doesn't leave rounded grooves in the teeth because the betel nuts are sliced up and mixed with a substance that enhances the release of stimulatory compounds, including one called arecoline.

Brumm wondered whether simply sucking on whole betel nuts was a precursor to this modern practice. The researchers soaked whole betel nuts in artificial saliva and added the resulting liquid to cells engineered with human brain receptors. That confirmed this behaviour would release enough arecoline to have physiological effects. They also found traces of arecoline on the ancient teeth, reinforcing their hypothesis (*Science Advances*, doi.org/hch528).

Arecoline has both stimulatory and pain-relieving effects. Brumm thinks that people may have started sucking on betel nuts to relieve tooth pain, but then became addicted. ■

Space

Alien oceans could one day be visible to us

Jonathan O'Callaghan

A NASA telescope in early development should be able to detect light bouncing off the surface of oceans on alien planets, vital information in the hunt for extraterrestrial life.

The Habitable Worlds Observatory is earmarked for launch in the 2040s. It will sport an advanced coronagraph, which can block out much of the light emanating from other stars. This will allow us to take images of potentially habitable Earth-like worlds outside our solar system for the first time.

Eleanor Cornish and Tyler Robinson at the University of Arizona have assessed whether the telescope will be able to spot oceans on these planets. Even though its images will be basic, with each world appearing as a single pixel, the pair found that light reflecting off the oceans – known as glint – should be noticeable when the planet is at certain angles around its star.

"Glint signatures can indicate the presence of liquid water on those planets, which is exciting because if we find water, that

It may be possible to see light glinting off an alien ocean



MARK GARLICK/SCIENCE PHOTO LIBRARY

suggests the planet could be habitable," says Cornish.

The study shows the glint will be detectable when the angle of the planet relative to its star and Earth, known as its phase angle, is greater than 120 degrees – meaning the light will bounce off the planet towards Earth, rather than out into space (arXiv,

"Clouds, vegetation or even rainbows might affect the light that is reflected to Earth"

doi.org/rnxb). "If it's at the optimal geometry, we should be able to see it," says Cornish.

The glint will cause the brightness of the world to rise slightly. How much the planet brightens in its orbit should tell us how much of the surface is covered by water, revealing if it is a water world entirely encased in an ocean or partially covered by land like Earth.

Sophia Vaughan at the Max Planck Institute for Astronomy in Germany says it might not just be oceans that the Habitable Worlds Observatory can detect. Other features on the planet, such as clouds, vegetation or even rainbows, might affect the nature of the light that is reflected to Earth. A rainbow would slightly focus the light of the planet, she says, "and you can possibly see that".

All of this information will be crucial to astronomers as they finalise designs for the telescope. "The fact that, maybe in my lifetime, we'll see the glint of an alien ocean is just incredible," says Vaughan. "I really hope that we do see it one day." ■

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Hear Nathan Mayne discuss the search for exoplanets and alien life on 11 October
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An AI experiment After months covering the rise of AI mathematics, **Matthew Sparkes** decided to put a free chatbot to the test – and was stunned at the speed at which it solved an outstanding problem



Matthew Sparkes is a reporter at *New Scientist* who covers robotics, quantum computing, AI and mathematics. He has a degree in AI and computer science

Matthew's week

What I'm reading

I am reading an extremely tatty copy of Len Deighton's 1966 spy thriller Billion-Dollar Brain, which fittingly features an all-knowing supercomputer

What I'm watching

Eccentric engineer Tim Hunkin's wonderful YouTube mini-lectures on the secret life of components. Only he could make an explanation of solenoids so interesting and quirky

What I'm working on

I'm delving into plug-in solar panels, which are now legal to use in the UK – but which some experts suggest need better regulation

LAST month, I interviewed mathematicians who were seeking a very peculiar set of dice. For no reason other than that they found the problem fun, they had hunted for five dice that could be thrown to decide who moves first in a board game. Crucially, each dice must have the same odds of winning and throws must never tie. Their search took more than a decade, but they finally cracked it.

I have also spent a lot of time reporting on the growing power of artificial intelligence to solve mathematical problems. The two ideas began to commingle in my head after I got off the phone with the dice hunters. Could AI go one better and find a solution for six dice?

So, I got up ChatGPT, described the problem, and asked for a solution. I nudged and clarified twice, and in total it “thought” for around 1.3 minutes. Then it delivered.

I promptly fired the solution off to Eric Harshbarger at Auburn University, Alabama, one of the researchers working on the five-dice problem. “I checked your numbers,” he says. They were correct.

I accept my solution wasn't a particularly elegant or aesthetically

“We're really going to be able to raise our ambition, specifically in mathematics, but in science in general”

pleasing one. Five of my dice had 720 sides; another had 20. Try rolling a 720-sided dice in the real world, or convincing players that a lopsided set is fair.

Nor was my AI-derived solution even the best we know of. Harshbarger and his colleagues have already been able to adapt their five-dice solution to create a six-dice one, where all dice have 360 sides, they told me. This isn't good enough for a public announcement and still



PCA PHOTO/SHUTTERSTOCK

When prompted, ChatGPT solved a decade-long maths problem about six dice in just 1.3 minutes

isn't very practical, although it is undeniably much better than mine.

However, a layman like myself found a solution to a problem in minutes by doing nothing more than describing it in plain English to an AI model. Something fundamental in mathematics has shifted, and new models, such as GPT-6, are emerging all the time with ever-better benchmark scores.

When I passed on the news, Paul Meyer, a software engineer at Google and another dice researcher who collaborates with Harshbarger, wondered whether AI had simply found the solution on the internet. But a search revealed no trace.

The best explanation that Harshbarger, Meyer and I came up with on a call is that the five-dice solution is published on the website the researchers use to track their work, as is a description of a method they have long known can “induct” a solution for more dice from one with fewer. Perhaps the AI model found these parts, filled in the gaps and did its own work.

“I think it's a big deal,” says Harshbarger. “A non-math person, at least, if they read those two components, might not be able to do [that] on their own. I hate

to say there's intelligence there, but it may have figured something out.”

Meyer is even more optimistic. “I do believe there's some intelligence there, personally. I'm almost constantly impressed – surprised and impressed – by what it can do on a regular basis, because it's come so far just in this last year, or six months.”

But in a strange sort of way, I am almost embarrassed to have done it. It took no skill on my part. I also feel it somehow undermines the work of the human mathematicians I had reported on and I fear it may be the thin end of the wedge for diminished human input in the field. What happens to pet mathematical ideas when a free online chatbot can chip away at them? Is this how future mathematicians, physicists and materials scientists will feel when their paper on groundbreaking AI-derived research is published?

Possibly not. “I would be thrilled,” says Harshbarger. “I don't care how a solution is found. I would certainly like to know, if we could, how the AI did it.”

Speaking to Sébastien Bubeck at OpenAI before the company's announcement that it had solved the Navier-Stokes Millennium problem (see page 4), I told him what I had just done with one of his models, and he was – with hindsight, unsurprisingly – seemingly neither surprised nor impressed.

“I feel we're really going to be able to raise our ambition, specifically in mathematics, but in science in general,” said Bubeck.

Bubeck conceded that things will change and this may cause some upset. The enjoyment derived from manually picking away at a problem and finding a solution may be a thing of the past. The community will also have to grapple with thorny problems like whether or not they can take credit for work AI does. Bubeck said mathematicians may be forgiven for a period of grieving. ■

The small things Research shows that short practices of mindfulness are as good as, if not better than, long ones. **David Robson** takes a look at how we can incorporate these microhabits into our daily lives



David Robson is an award-winning science writer specialising in the extremes of the human brain, body and behaviour. His latest book is *The Laws of Connection: 13 social strategies that will transform your life*

David's week

What I'm reading

Besides *Micropractice*, I've been enjoying the audiobook of *The Odyssey*, translated by Emily Wilson and narrated by Claire Danes

What I'm watching

I've been binging the new season of *Ludwig* from the BBC

What I'm working on

I'm writing up a feature about consciousness in humans, animals and AI – more soon!

LIKE many people, I have the intention of practising self-care. I know that meditating each day will help break me out of stressful thought spirals, but it has never become a part of my regular routine. With life commitments, there just doesn't seem to be the time.

A new book – *Micropractice: A science-backed approach for calm, clarity and joy in 30 seconds or less* by psychologist Eli Susman – might offer the solution. Drawing on a growing body of research, he argues that we can glean many of mindfulness's benefits from exercises lasting 30 seconds or less.

I would have been sceptical if I hadn't previously written about some of Susman's research. A few years ago, he assessed the benefits of a very short self-compassion meditation, which encouraged participants to "send kindness and warmth" to themselves "by bringing one hand to your belly and the other to your chest, with the energy of giving yourself a hug". The aim was for participants to treat themselves like they would treat a friend.

That was it – 20 seconds of effort to treat themselves kindly. They were then asked to practise it daily for a month. Those who followed the instructions showed considerably less stress, and better overall mental health at the follow-up

Taking brief moments each hour to stretch at work could help boost your wellbeing

compared with people who had instead practised a short sequence of finger-tapping exercises.

In his new book, Susman describes similarly optimistic results for many other interventions. "Study after study pointed to the same overlooked pattern – one that almost no one talks about: It's not

"Even a short period of bodily movement can bring enormous benefits for our mental state"

how long you practice, but how often," he writes. "Consistency, not duration, is what drives change."

Consider "breathwork", which typically involves slowing our inhales and exhales and has proven benefits for our wellbeing. Intervention lengths vary widely, but a large meta-analysis suggests that the duration is largely unimportant. "Although slow breathing exercises significantly reduced stress, depression, and anxiety, the size of the effects didn't change based on how long participants practiced," writes Susman.

The same goes for stretching and yoga poses. Even a short period of bodily movement can bring enormous benefits for our mental state. As evidence, Susman points to a study of surgeons who were asked to perform stretches for 90 seconds every 40 to 60 minutes

of the working day. The minimal adjustment to their routine improved their mental focus and reduced the physical pain that normally accompanies their work.

The brevity of exercises you could attempt should make them easier to fit into our busy lives, but that is no guarantee that they will become routine. For this reason, Susman has analysed the available literature on behavioural change to build a brief course on habit formation.

The trick, he says, is to first find a cue that anchors the habit in a specific event. Ideally, the cue should be something conspicuous that occurs very regularly. As a coffee addict, I might practise the self-compassion meditation while I wait for my espresso to pour, for instance.

According to Susman, you should enhance the sense of reward by articulating why you want to change your behaviour in the first place and savouring those benefits afterwards. As you get started, you should also keep track of your progress, which can help to enhance your sense of accountability.

Most counterintuitively of all, Susman suggests allowing yourself two "emergency skip days" each week, during which you can cut yourself some slack if you fail to meet your goal. "Because what derails progress isn't missing a day – it's the shame spiral that follows."

In a forthcoming study, Susman found that teaching these simple tips in a 15-minute video vastly increased the chances that someone would practise the self-compassion meditation over the following six months. And that, in turn, determined the size of the benefits for their mental health.

Susman's argument is strong enough that I have started adding a few of these suggestions into my life, and I am already appreciating the moments of calm that they bring. I can only hope that this continues in my most hectic periods, too. ■



WESTENDG/DEEPOL BY PLAINPICTURE

Cosmic mirror It is only natural to view the cosmos through the lens of humanity – but we can be in awe of the universe without inserting ourselves as a main character, says **Chanda Prescod-Weinstein**



ADRIENNE BRESNAHAN/GETTY IMAGES

AS A science writer, I spend a lot of time worrying about how to make my words simultaneously accurate and accessible. Most of the time, that's things like correctly explaining the concept of space-time – the idea that space and time are a single, joint phenomenon – without using any equations.

In the work of cosmology, there is also the element of awe. All my training teaches me to try to awe my audience. Look at the cosmos in all its glory! But the political pressures I face also teach me that I need to explain why looking at the cosmos in all its glory is practically useful. Suddenly, I feel responsible for making the case that it's valuable to humanity to know things about the cosmos.

There are many ways into this. For example, we often hear that we are carbon-based life forms. Carbon exists because compact balls of hydrogen heated up and became dense enough for nuclear fusion to begin. Which is to say, stars formed, and carbon is a by-product of a star. "The cosmos is within us. We are made of star stuff," the astronomer Carl Sagan famously said.

Astronomer Carl Sagan famously said that "the cosmos is within us"

There is a kind of risk of looking at the cosmos through ourselves, though. Queen Victoria's poet laureate Alfred Tennyson noticed this early on while he was an undergraduate at Trinity College, Cambridge. While there, Tennyson composed a bitingly satirical poem, making fun of another student, *A Character*, which begins:

*With a half-glance upon the sky
At night he said, 'The wanderings
Of this most intricate infinite
Universe
Teach me the nothingness of
things.'*

On my first careless read, I thought this was a nice entry point to the cosmos, this most intricate infinite universe, which teaches us how small and fleeting we are. I didn't notice "With a half-glance", which suggests the looker is doing so carelessly. I only realised the direction the poem was going in when I got a few more lines in:

*He spake of beauty: that the dull
Saw no divinity in grass,
Life in dead stones, or spirit in air;*

As I read these three lines, I found myself nodding. I have crossed paths with people who see our planetary home only as something to pillage, rather than a wonder to honour.

Then the poem takes a turn:
*Then looking as 'twere in a glass,
He smooth'd his chin and sleek'd
his hair,
And said the earth was beautiful.*

Here Tennyson acerbically attacks his subject as completely vain, treating the cosmos like its meaning is entirely as a looking glass for himself. In the process of saying the world is beautiful, he really means himself.

I should have seen it coming. He gave the universe only "a half-glance" but then claims to know so much about it. I first read this part of the poem in *The Boundless Deep: Young Tennyson, science, and the crisis of belief* by Richard Holmes, a biography of Tennyson's early years. In the margins next to it, I made a note: scientists and glory. In my notebook I continued: Astronomers as cosmic narcissists. What happens when we ask the cosmos to make us great?

These thoughts emerged from my experience as a scientist who is constantly under pressure to prove that my contributions aren't just valuable but also noteworthy.

In a system that trains us to extract value for ourselves from the cosmos, we forget that its beauty can be the justification. That the sense of connection it generates between us might be enough. The man in the poem takes a half-glance and insists that the universe teaches us the nothingness of things. But when we look more fully, we see that the universe is plentiful and so complex that studying it is a multi-generational effort, one where we work together to understand something bigger than ourselves. ■



Chanda Prescod-Weinstein is an associate professor of physics and astronomy at the University of New Hampshire. She is the author of *The Disordered Cosmos*. Her latest book is *The Edge of Space-Time: Particles, poetry, and the cosmic dream boogie*

Chanda's week

What I'm reading

Obviously enjoying The Boundless Deep: Young Tennyson, science, and the crisis of belief by Richard Holmes

What I'm watching

Silo series 3 has me in full conspiracy-theory mode

What I'm working on

I tried taking a vacation, but spent a lot of it worrying about the multinational science funding crisis

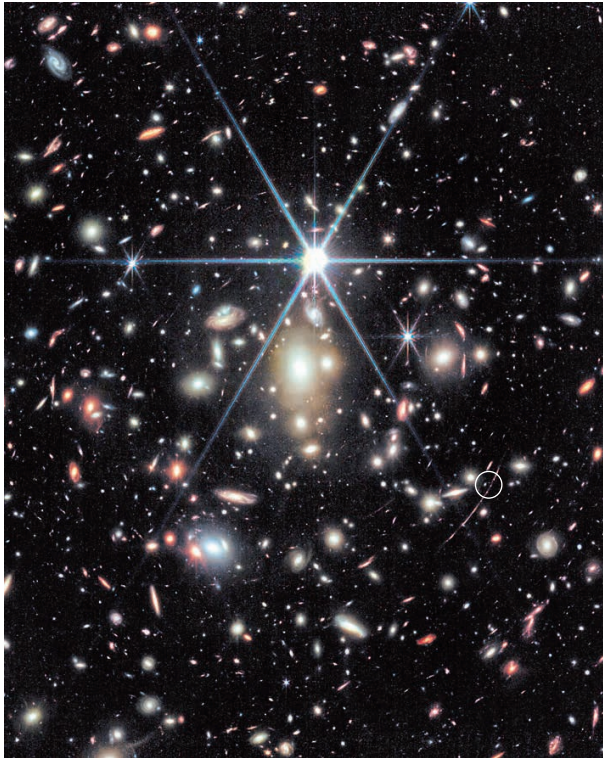


Hear Chanda talk about how dark matter and dark energy shape our cosmos on October 10
[newscientist.com/nsimag](https://www.newscientist.com/nsimag)

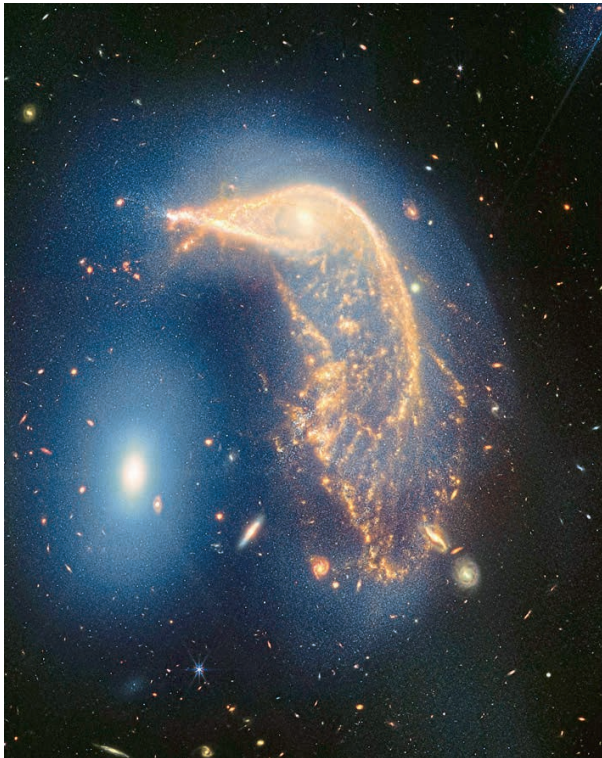
NASA, ESA, AND A. LEROY (THE OHIO STATE UNIVERSITY)



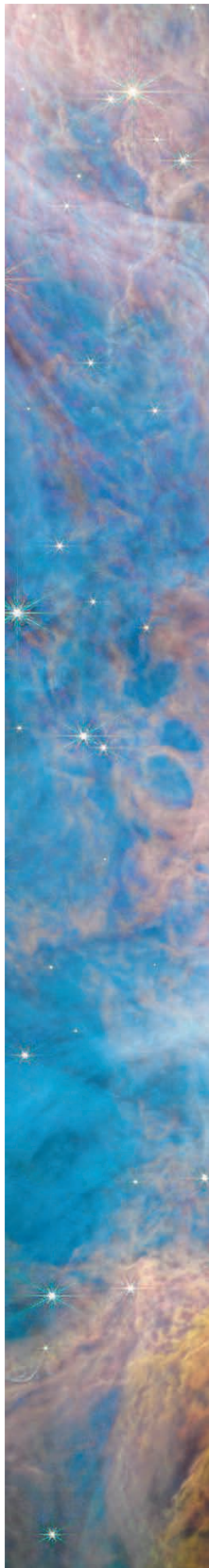
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NASA, ESA, CSA, STSCI



ESA/WEBB, NASA, CSA, M. ZAMANI (ESA/WEBB), THE PDR4ALL ERS TEAM





Ancient cosmos



Michael O'Mara Books

THE most powerful telescope ever launched into space is also a time machine. Ever since it left Earth in 2021, the James Webb Space Telescope (JWST) has been helping researchers better understand our cosmos by pointing its several powerful instruments at our galactic neighbourhood and way beyond.

In *James Webb's Ancient Cosmos: A journey to the beginning of time*, co-written with Sarah Wild, space scientist Maggie Aderin collects nearly 200 images obtained by the telescope, illustrating how it is also looking backwards in time.

"We call it a time machine because it has the capability to look into the distance," she says. "We're looking at objects which are sometimes over a billion light years away. Because it takes that light so long to reach us, we are looking at the ancient universe."

At a distance of 68 million light years is the barred spiralling M58 galaxy (upper far left). In the lower far left is the most distant star humanity has ever seen. The light from this star, Earendel, has been travelling towards us since the universe was about 1 billion years old. It appears in the image as a fuzzy white dot within the circle.

The image to its right shows a pair of galaxies that collided in the past and now resemble a penguin-and-egg pair, 326 million light years from Earth. The near-left image shows the Orion Bar region, an area of star formation and intense activity, ripe for study.

"About 10,000 scientists and engineers from all around the world collaborated on this; I love that global unity. To understand the universe, we have to work together," says Aderin. ■

Karmela Padavic-Callaghan

Of physics and philosophy

In an impressively broad new book, Carlo Rovelli shares how he sees physics shaping our world. **Karmela Padavic-Callaghan** finds it compelling



Book
On the Equality of All Things
 Carlo Rovelli
 Translated by Simon Carnell
 Allen Lane

AS THE science that describes the ingredients of our reality and how they interact, physics can make a strong claim to helping us understand our surroundings. But what about the making of meaning more broadly?

In *On the Equality of All Things: Physics and philosophy*, physicist Carlo Rovelli tackles this question to great effect. He argues that, with the advent of quantum physics, science underwent a revolution, but philosophers haven't fully caught up. Writing with a spirit of enthusiastic defiance and ardent conviction, Rovelli aims to offer them (and us) a crash course.

He is no stranger to ranging across physics, philosophy and the history of science, and has written very popular books on quantum physics, its history and its myths, such as *Seven Brief Lessons on Physics*. As a physicist,

he has worked on quantum gravity, or how to harmoniously combine quantum mechanics and Albert Einstein's theory of general relativity – to date, two of the most successful branches of physics.

Rovelli is also writing from the perspective of an interpretation of quantum theory he pioneered: relational quantum mechanics, the idea that objects don't exist independently, but in relation to each other. He uses this as a starting point for a

“For Rovelli, quests for absolute truths are projections of psychological and cultural anxieties”

bigger discussion he developed as lectures to philosophers at Princeton University. This is why his new book is structured in a pedagogical way, helping readers follow his arguments.

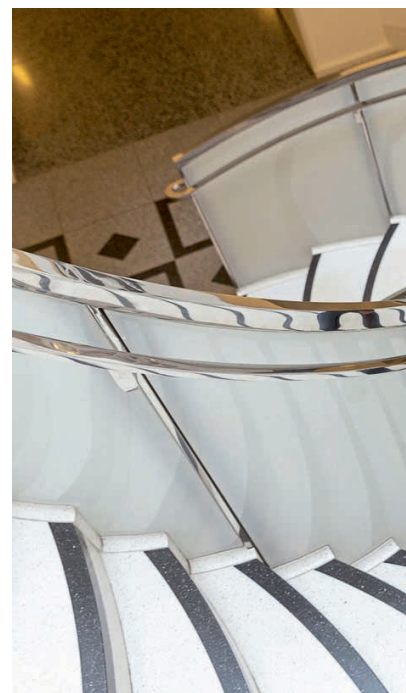
From the outset, Rovelli carefully instructs readers to recognise implicit assumptions they hold about the nature of our world, ones that depend on our

specific place in the universe. For example, because Earth is fairly small, we have a shared notion of time, which wouldn't be the case if we had grown up playing phone tag with friends across the galaxy.

The same is true of how we conceptualise things like truth. “Adjectives like ultimate, truly, real and objective turn out to be meaningless,” he says, establishing one of the book's themes. For Rovelli, quests for absolute truths are projections of psychological and cultural anxieties.

This dovetails nicely with relational quantum mechanics, where the key ingredients of reality aren't objects, but relations and processes. Rovelli doesn't worry about knowing an object for certain, so long as he can describe how it interacts with, or changes in relation to, other objects.

What could seem like empty relativism is made concrete with the help of quantum physics, and Rovelli anchors his ideas by insisting that experiences – also relational – ought to matter more than abstractions. “Scientific models are conditional statements,” he says, underlining

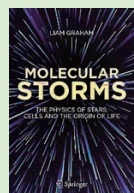


that even the best theories can change if confronted by new data.

Absolute certainty, strict causality and objectivity are all idealisations that can hold science and philosophy back when applied too strenuously, writes Rovelli. There is no space or time in his world, just quantum processes: space and time are things we can name because they eventually touch us. “The world is therefore relative information between its different parts,” he states. “The causal orientation of the world is only statistical.”

In the book's second half, Rovelli uses ideas of information, dissipation and entropy to round out his picture of the world. Many of these ideas aren't new. For example, the notion that much of our world is determined by existing in an entropy gradient – a state where entropy is increasing, or moving from a few tightly ordered configurations to a mess

Three more books on how we make meaning



Molecular Storms:
 The physics
 of stars,
 cells and the
 origin of life
 by Liam Graham

Liam Graham, a physicist turned economist, has produced a clear, unfussy exploration of how ideas about statistics, ensembles of objects and entropy are crucial for understanding our world.



The Blind Spot:
 Why science
 cannot ignore
 human experience
 by Adam Frank,
 Marcelo Gleiser
 and Evan
 Thompson

A physicist and two philosophers decisively show how any view of science that moves away from the human relation to the world and up the “spiral of abstraction” is bound to fail – or lead to harm.



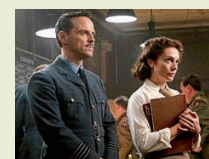
*Meeting the
 Universe Halfway:*
 Quantum
 physics and the
 entanglement
 of matter and
 meaning by
 Karen Barad

This is an academic book that takes some effort to parse, but also a masterclass in how the philosophy of early quantum theory can be related to ideas from critical and feminist theory.



Catherine de Lange
Editor
London

My grandfather was the commander of a minesweeper in the British navy during the second world war. The night before D-Day, he wrote in his diary: "And now maybe we are going



to do the big do. Maybe I won't come back. A few words in case I don't, to tell you what I was..."

Originally planned for 5 June 1944, the D-Day landings were delayed by a day due to the weather. The drama and jeopardy involved are retold in a gripping new war drama, **Pressure**. It recounts the story of meteorologist James Stagg, played brilliantly by Andrew Scott (above, left), who is brought in to give the all-clear on the weather for the invasion. Stagg predicts building pressure that indicates devastating storms, which would doom the operation. His US counterpart dismisses this with ridicule, insisting the weather will be fine.

I don't normally seek out war movies, but this was tender and clever, yet still pacy. I only wish I could discuss it with my grandfather, who did survive D-Day, in no small part thanks to Stagg.

ALEX BAILEY/FOCUS FEATURES/STUDIOCANAL



**There is no space or time
in Carlo Rovelli's world,
just quantum processes**

insistence on relationships, his allergy to absolutes and his humility about how often experience teaches us to change our scientific minds compelling.

Towards the end of the book, he writes that the world is our "blood brother", that "We face the world not as an 'I' faced with a 'this', but as a 'me' faced with a 'you'. We are in the world. We are here. And we are home." Seeing ourselves as abstract consciousnesses disconnected from the world and looking for truths beyond how it touches us is a fool's errand. Here, Rovelli and I are in harmony.

Yet there are imperfections in Rovelli's argument. In arguing against contemporary philosophers who advocate a more absolute view of the world, many of his citations point to decades-old philosophical works. Rovelli is also quick to call arguments and ideas "nonsense" – which is impactful and charming in a lecture, but, page after page, can seem condescending.

And in discussing modern science, misconceptions of how it ought to work and what its goals are, I kept waiting to read about the societal forces shaping science. How such an expansive book can fail to mention political and economic forces that influence and benefit from some of the philosophical beliefs Rovelli attacks baffles me a little.

Ultimately, though, *On the Equality of All Things* is a worthwhile intervention in modern science and philosophy, and because Rovelli's writing flows smoothly between difficult concepts, it is fairly accessible and rich with footnotes for all who want to know more. ■

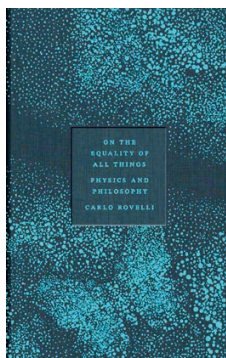
of many equally likely ones – will be familiar to followers of modern physics. But, in Rovelli's view, philosophers haven't taken many of these ideas seriously enough.

On the Equality of All Things is impressively broad, and Rovelli moves easily from quantum gravity and entropy to free will, consciousness, morality and how science works. Throughout, he extrapolates from ideas developed in physics. "Knowledge," he writes, "is primarily a physical correlation", again harking back to relational quantum mechanics.

From this, it follows that all our knowledge is contextual and perfect certainty is meaningless: there is no secret foundation of reality beyond what we can access through physical processes. The world can be explained by physics, and physics offers us only relationality, perspectivism and, at times, vagueness. If this offends us, that is a function of culture, not a

problem with the world, he argues.

Given my traditional physics training, I bristled at some of Rovelli's ideas – even if they didn't seem so troubling after a few more moments' thought. This may be a testament to how willing he is to challenge the orthodoxy people like me absorbed at university. As a writer who has tried to describe the lessons that quantum physics offered my own life, I found his



DAVID STOCK

Across worlds In a near-future Los Angeles just out of curfew, two women attend a party that gets stranger by the minute. Welcome to the parallel universes of *Station Eleven* author Emily St. John Mandel's latest, says sci-fi columnist **Emily H. Wilson**



Emily H. Wilson is the author of the *Sumerians* series (*Inanna*, *Gilgamesh* and *Ninshubar*, all published by Titan) and she is currently working on her first sci-fi novel. She is a former editor of *New Scientist* and you can follow her on Instagram @emilyhwilson1



Book

Exit Party

Emily St. John Mandel
Pan Macmillan

Emily also recommends...

TV

Fringe

Created by J. J. Abrams, Alex Kurtzman and Roberto Orci
This five-season TV show was uneven on many levels, to say the least. However it was also a bold and at times wildly creative exploration of parallel universes, with moments of absolute genius.



BRUNO PASSAGI/TIGETTY IMAGES

IT IS always a big deal when Emily St. John Mandel, the much-feted author of *Station Eleven* and *Sea of Tranquillity*, has a new novel out, and her latest doesn't disappoint.

Exit Party is a parallel-universe story that flits backwards and forwards in time, from character to character, and eventually between worlds. It is as elegant and brilliantly executed as her fans have grown to expect, a book where every detail matters. I will give only the general set-up in this review, so no need to fear spoilers.

We begin in California, 2031. The US has been torn apart by war, but the curfew in Los Angeles has just been lifted. Our main protagonist, Ari, and her roommate Gloria plan to go to a party. (I read this very early section a few times before sticking with it, so if you find yourself in the same situation, know that it's worth pushing on.)

Ari has just spent 20 years in prison for taking out a hit on someone, and she is very keen to party, but Gloria is reluctant to leave home when it comes to it. She shows signs of PTSD and mentions another party where

a bomb went off. Eventually the two get to the party, and find the house full of flowers and the host, Kareem, warm and welcoming. At first, everything feels magical and exciting, but later something goes badly wrong, in ways that will take years for Ari to unravel.

"There were extra people," is how a guest at the party explains it. Yes,

"Exit Party is as elegant and brilliantly executed as fans have grown to expect"

there are too many people at the party. At one point, Ari realises a woman is there in two completely different outfits; are they twins? The next day, Kareem is missing, having vanished entirely.

Exit Party's structure is unusual, playing with time and points of view as your understanding of what is really going on deepens. St. John Mandel's plotting skills and ability to build character quickly mean that the book never loses its dramatic intensity and

Who were the extra people at what seemed like a great party?

forward propulsion as we make our leaps in time and viewpoint.

Thinking over the book, once I had finished it, I was struck by how much St. John Mandel was able to explore through her two-worlds structure, in what isn't a long book. We get to imagine how different a person might be if life dealt them a different hand, or if they lived in a different world, a world also broken but in a very different way, perhaps. We also get to ponder how, without witnesses to sections of our life, we can end up in a different world without having to cross between universes. Oh, and so much more. It is a wonderfully thoughtful book.

Novels by St. John Mandel are usually described as speculative or dystopian fiction. This one also counts as old-fashioned sci-fi, I guess, because of the parallel universes, but there is close to zero science, because how people cross between worlds isn't the point. This is about fate, courage, character and destiny, through the prism of parallel worlds.

A key facet of the book is also that Ari has premonitions, where she (sort of) knows what is going to happen next. Again, a science-free zone, but it makes for great entertainment.

My only bone to pick is that I never bought Ari's full backstory. We are meant to believe that this canny, well-prepared, criminally savvy and basically OK woman, who also happens to experience strong premonitions, would take out a clumsy hit on one of her very few close relatives. Hmmmm.

Anyway, that aside, bravo to Emily St. John Mandel, an absolute force in modern fiction. ■

Term-time working.

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Keep us forever without borders



HENRY GRAY HAS WORKED FOR MSF IN NUMEROUS CRISES AROUND THE WORLD.

"I've worked for MSF on and off for twenty years, but I still remember the first project I worked on. We were setting up a hospital in the middle of a forest in the Democratic Republic of Congo. There had been a lot of fighting in this area, and although the conflict had died down, the legacy of war was still very visible. About 125,000 people lived in the immediate area, but there were no services, and not even basic medical care was available. Women couldn't get caesareans if they needed them; there were no medicines, no vaccinations, and no treatment for diseases. It was a desperate situation, and that's when MSF sent us to get an emergency hospital up and running.

From the start, our goal was to provide medical care and save lives. But we knew that for this work to continue and to have lasting impact, we couldn't just provide medical care. We needed to bring the community on board; we needed to hire local staff and train local people to do the jobs that we were doing. So that's what we did.

It was fantastic to see the immediate impact of that lifesaving work. But what really surprised me was seeing how the hospital transformed the entire area. I still remember the day we switched on the generators that would power the lights in and around the hospital. Suddenly, schoolchildren started turning up to sit under the lights and do their homework. Over the weeks, businesses sprang up, and the local economy grew.

Over the years I've worked in war zones, in Ebola outbreaks and in natural disasters, and I've seen over and over again how the simple actions we take as MSF can have massive and positive knock-on effects for people, for their families and their communities.

But none of those simple acts happens without your support. MSF doesn't take money from governments. We're privately funded by people just like you. By leaving a gift in your will to MSF, you can know that your gift will not only save lives, but will change lives now and into the future. Thank you."

Above 4-year-old Franyeli receives medical care at MSF's mobile clinic in Puerto Alegre, Colombia. Photo: © Natalia Romero Peñuela/MSF

"We will always be committed to providing care to those who need it. But that legacy is not ours alone."

Dr Rachael Craven,
Anaesthetist and President



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Editor's pick

We need more research on female hormones' effects

22 August, p 28

From Alex Spencer,
Burwell, Cambridgeshire, UK

I was very interested to read your feature on how female hormones affect ADHD. I would love you to take a deeper dive into a subject that many women are only learning about from each other: progesterone intolerance.

Social media platforms and online forums discussing hormone replacement therapy and hormonal contraception are full of women desperate to feel the benefits of these medications, but when they try them, some discover that progesterone plunges them into the bleakest, darkest depression of their lives. In such treatments, you have to take progesterone with oestrogen to avoid uterine cancer, but when women tell their doctor that their mood is affected, they are often met with a shrug. The only options on offer are lower-dose hormonal coils or taking progesterone vaginally. This doesn't help many women.

I would love to see some proper, rigorous investigation into this subject, as all the "menopause experts" seem to skate over it.

The implications of getting rid of irrational numbers

29 August, p 30

From Carl Zetie,
Raleigh, North Carolina, US
Physicist Tim Palmer wants to banish irrational numbers from quantum mechanics, but why stop there? Everything we can know about the physical world is ultimately based on a measurement, and any quantity we can measure is either a fraction or a multiple of whatever units we use for our rulers, clocks and scales. We might imagine a 1-metre square and calculate that its diagonal should be $\sqrt{2}$, but when we come to measure it, we will only ever find a rational

number – unless our ruler is subdivided with infinite precision.

So is it truly meaningful to say that quantities that appear only in our equations, but never in our measurements, physically exist? Perhaps, in some way analogous to Palmer's rational quantum mechanics, irrational quantities might appear in the maths, but be excluded "states" in the physical universe.

From Ian Roselman,
Leighton Buzzard, Bedfordshire, UK
I find myself hoping that rational quantum mechanics turns out to be right. Could it be that numbers themselves are quantised if they are to have any meaning in reality?

I notice that the estimated maximum number of states of a qubit is 2^{400} , and this number is close to 10^{120} . This is the scale of the biggest discrepancy in physics: the vacuum catastrophe, the mismatch between the predicted energy of empty space and the amount we actually observe. It is also the result of my simplistic estimate of the number of subatomic particles (real or virtual) that would fit into the known universe. Coincidence? Maybe, but it is fun to speculate.

Key scientific tool may be impaired by solar storms

15 August, p 34

From Bryn Glover, Kirkby Malzeard,
North Yorkshire, UK

As you report, solar storms are injecting random quantities of carbon-14 into our atmosphere to the point where spikes are recorded in tree rings. So, to what degree can we rely on carbon-14 dating to determine the age of organic material, given it theoretically depends on a constant supply of the isotope?



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More proof needed for AI mathematicians

15 August, p 38

From Andrew Shead,
Tulsa, Oklahoma, US

Terence Tao discusses the recent glut of AI-written proofs of mathematical problems. I would argue that such proofs aren't true until validated by human mathematicians. Just as a child's schoolwork doesn't pass unless they show their working, so an AI proof can't pass without showing its step-by-step process.

Validating proofs is one of the roles of the mathematician. Doing so means making it understandable in stages, which involves a combination of human linguistic explanation, mathematical equations and Lean programming code, with all logical leaps eliminated. Mathematicians must force AI to show its workings, with full citation of the prior work on which it depended.

A fish-friendly way to get omega-3 in your diet

Letters, 22 August

From Erik Foxcroft,

St Albans, Hertfordshire, UK
Sam Edge is right to point out that eating fish to get more omega-3 can be harmful to the environment if they aren't sustainably sourced, and that there are many alternative plant and fungus sources. However, if you want to make sure you are getting enough, you should know that the fish don't make omega-3 themselves, but get it from eating algae.

Some supplement producers harvest these algae directly and put them into capsules, which look like cod liver oil, but don't involve killing fish.

Two ways to ensure more action on climate change

Letters, 29 August

From David Flint, London, UK
Peter Brown calls for a global effort to create disincentives for climate-damaging activities, as well as subsidies for the mitigation of adverse impacts. The need for this has been obvious for 30 years, but attempts to take action have been sabotaged by fossil fuel giants and petrostates.

We won't get the sustainable economy that Peter and I – and, I think, many other *New Scientist* readers – want unless we call this out and support parties and politicians who will do what is right for Earth, our shared home.

From Simon Evans,

Malvern, Worcestershire, UK
I agree that an out-of-control market economy is responsible for the climate crisis, but I doubt that global markets will be reined in any time soon. Our instinctive urge to compete and to acquire wealth and power, felt at all levels from individual to international, has made our world what it is, both good and bad.

Our amazing ability to innovate and make things happen needs to be channelled towards efforts to reverse the effects of climate change, such as geoengineering. This plays to our strengths: we are very good at doing things and very poor at not doing things.

To deer, cougar presence is claws for concern

15 August, p 13

From Eric Kvaalen,

Les Essarts le Roi, France
Liana Zanette says the presence of cougars creates a "landscape of fear", and this "is good news all around". Tell that to the deer. ■

For the record

■ Gravity affects objects with mass or energy (29 August, p 30).



AHRA KWON

PHYSICISTS aren't known for playing fast and loose with the rules of reality. But when it comes to the concept of space, some take a remarkably permissive attitude. For most of us, it would seem there are just three dimensions of space. Yet string theory, for instance, postulates there are at least nine.

The same attitudes don't tend to apply to time. For almost all physicists, time has just one dimension, period. Yet some dare to wonder if there might be more to time than meets the eye.

It is a dangerous game – mess with time, and you can end up scrambling the laws of cause and effect. But then again, we know physics has plenty of problems. Maybe a drastic step is exactly what is needed to solve them? “Everything in reality can come out from a theory with two times,” says Itzhak Bars at the University of Southern California, who has been wondering about this for decades.

With his ideas and those of others now hinting at what might lie inside black holes and even suggesting a possible solution to one of the biggest mysteries in quantum mechanics, perhaps this wild idea deserves the time of day.

The edifice of modern physics is supported by two great pillars. Quantum theory describes

the world of particles and three of the four fundamental forces, while general relativity explains the remaining force, gravity, as the curvature of space-time. Yet nearly every deep problem in physics is a crack along the same fault line: these theories paint incompatible pictures of how reality works.

Broadly speaking, physicists suspect a deeper picture underlies both theories. Trying to work out what that is has led them to ideas like string theory, with its many extra dimensions of space. But few think we are anywhere near the true answers yet.

So, what about time, then? At first blush, Albert Einstein's ideas about relativity do at least teach us that time is far stranger than everyday experience suggests. We know that distances and durations aren't fixed properties of the world, but depend on how fast you are moving. Two observers whizzing at different velocities close to the speed of light can disagree about how far apart two events were and how much time elapsed between them, without either being wrong.

Even so, Einstein insisted they had to share an objective reality – which he hinged upon the speed of light. Every observer, he argued,

must measure light travelling at exactly the same speed in a vacuum, and for that to work, space and time couldn't behave independently. A change in one had to be compensated for by a change in the other: the faster you move through space, say, the slower you move through time.

The upshot is that you can't just bolt on another dimension of time without risking some pretty disastrous consequences. Add a second temporal dimension, and that compensation can break down; instead of keeping the theory in balance, the extra time contribution can lead to outlandish predictions such as negative energies and events that have a less-than-zero chance of happening. Physicists call such nonsense predictions “unphysical”, and they can be difficult to exorcise from ideas with more than one time.

Even worse, causality can suffer, too. With two temporal directions, there may be routes through time that let you evade the ordering imposed by the other. Events that seemed safely arranged into before and after can become connected in unexpected ways, creating loops or allowing an observer to bypass part of history. In principle, you could “go around any event you wanted to avoid”, says theoretical physicist Chris Hull at Imperial College London. “Biology as we know it would be very hard to envisage in such a theory.”

It is safe to say that messing with time isn't for the faint-hearted. But some are unfazed by the challenge – and no one has spent more, well, time thinking about these ideas than Bars. In 1998, he unveiled “two-time physics”, a framework with four dimensions of space and two of time.

Bars says the idea grew out of thinking about two of the most basic things about an object: its position and momentum; in other words, where it is and how it is moving. At first, the two seem quite different. One tells you where an object is; the other, what it is doing. But quantum mechanics hints that they are intimately linked. According to a foundational rule known as the uncertainty principle, you can never know both with arbitrary precision. Pin down an object's position more exactly, and its momentum becomes less certain, and vice versa.

Bars wondered whether the connection went deeper still. After all, knowing an object's momentum allows you to predict where it will be later. The history of physics, too, has vindicated many efforts to unify once-disparate ideas – electricity and magnetism ➤

Time's extra dimension

We have long thought that time had only one dimension, but what if we were wrong, asks **Miriam Frankel**

forming electromagnetism, for example.

By treating position and momentum as merely alternative ways of describing the same underlying reality, Bars predicted that reality could have extra dimensions, including a second dimension of time. One way to picture this strange second time is to imagine shining a light on a cylinder. From one angle, its shadow is a circle, yet from another, it is a rectangle. They may look completely different – the same way momentum and position do – yet both can come from the same object. “Think of four space and two time dimensions as a room, and our three space and one time [dimensions] as its walls. On each wall, you see a different shadow of the same reality,” says Bars.

Crucially, treating position and momentum as interchangeable descriptions of reality also avoids the problems that plague more naive attempts to add a second time. Once you choose one way of looking at the system – in terms of position rather than momentum, say – one space and one time dimension effectively drop out of the physics, leaving us with the familiar picture of space-time and eliminating the usual causal paradoxes.

This may all be mathematically workable, but that still leaves a more basic question: what does it actually buy us? “There needs to be a clear sense of where it’s going and what kind of progress it might lead to,” says Emily Adlam, a philosopher of physics at Chapman University in California.

In 2013, Bars began applying his ideas to cosmology with physicists Paul Steinhardt at Princeton University and Neil Turok, now at the University of Edinburgh in the UK. But before long, the work began producing some rather unsettling results.

Treating position and momentum as interchangeable may be a simple idea, but it has surprisingly far-reaching consequences. A quantity that looks fixed in one four-dimensional shadow may take on a completely different value in another. This means that things we normally regard as constants, such as particle masses or the strength of gravity, could potentially vary.

This is exactly what Bars and the others saw when they considered a universe collapsing towards a “big crunch”. The model predicted that particle masses dwindled towards zero as gravity became overwhelmingly strong. Then, instead of ending at the crunch, the universe passed through a brief phase in which gravity became repulsive, before emerging into another big bang.

“Everything in reality can come out from a theory with two times”

“When I first saw that, I was afraid. I thought maybe something is wrong with my theory,” says Bars. The result ruffled him enough that he largely put two-time physics aside for more than a decade – retreating instead to the comparative comforts of string theory and theoretical particle physics. “But finally, I decided that either I give up my theory or I accept that it says something interesting.”

In 2025, Bars returned to the idea and asked whether the same machinery could tackle a similar situation: the one inside a black hole. A universe collapsing into a big crunch and matter falling into a black hole both head towards points, known as singularities, where quantities such as density or the curvature of space-time shoot towards infinity and equations stop making useful predictions. Most researchers take that as a sign that the physics is incomplete. But Bars argues that a falling particle has no reason to care that our equations have failed. “Why stop?” he says. “Something should happen.”

In his six-dimensional picture, it does. Rather than merely ceasing to exist, a particle falling into a black hole keeps going. Echoing Bars’s earlier cosmological results, its path carries on through the point where ordinary physics breaks down and into another region of space-time on the far side. There, gravity



flips sign and becomes repulsive – an antigravity patch within the black hole.

It might sound wild, but Bars says it could offer a new angle on an old problem, known as the black hole information paradox. Quantum mechanics says information can't be destroyed. Yet Stephen Hawking, in the 1970s, showed that black holes should slowly radiate away their energy, apparently leaving no trace of anything that fell inside. In Bars's model, the information doesn't disappear. It merely passes through the singularity, emerging on the far side, where gravity becomes repulsive.

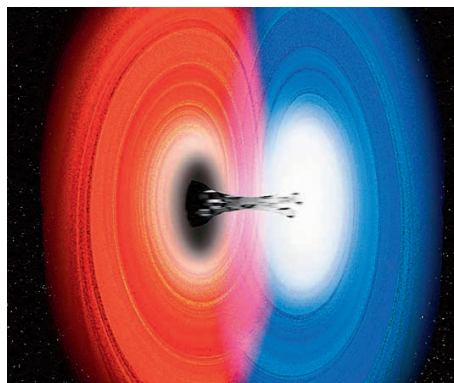
"The topic and techniques introduced in this paper are highly innovative," says Federica Muscolino, a theoretical physicist at the University of Milan-Bicocca in Italy. "It resolves the problems posed by singularities in the universe and provides a solution to the black hole information paradox."

Could we ever know if any of this is right? In principle, yes. Astronomers can look for signs that supposedly fixed quantities have changed over cosmic history. Two-time physics predicts that such "constants", including particle masses, could shift in extreme environments. Although finding such a variation wouldn't immediately prove Bars right, it could offer a rare glimpse of the hidden structure he thinks lies beneath ordinary physics.

However, one thing to remember about Bars's hypothesis is that the extra time dimension isn't something we can travel through or experience. That makes some physicists suspicious. "It's a nice, interesting idea. It's a nice way of looking at the equations," says Hull. "But it's mathematics rather than physics."

Even so, the idea of two times has got its hooks into other physicists. Marco Pettini at Aix-Marseille University in France is among those trying to introduce a more physical version of a second time without falling into its usual pitfalls. His route into the idea began with a conversation with Nobel laureate Roger Penrose about quantum entanglement, a strange effect linking the outcomes of measurements on two particles, even when enormous distances separate them. Measure one, and you can immediately infer something about its partner, even though no ordinary signal appears to pass between them.

That sits uneasily with locality, the principle that an object can be influenced only by what is nearby, or by something that has had time to reach it. Quantum mechanics predicts these correlations perfectly, but doesn't explain how the particles are linked.



In some versions of physics with two time dimensions, a black hole could contain an antigravity patch

"Entanglement is really astonishing because it has no place in our space-time," says Pettini. In 2025, he suggested taking that literally. If this type of connection couldn't be seen in our familiar space-time, perhaps it occupied another dimension. Extra space wouldn't help, he argued, because information would still need to travel through it. "So, the only possibility is the extra time dimension," he says.

Explaining entanglement

To picture what he means, imagine crossing London's Trafalgar Square, when everything around you suddenly freezes. While the rest of the world remains pinned in ordinary time, you slip into another temporal dimension and drag one of the bronze lions across the square. When normal time resumes, nobody will have seen the journey. The lion has simply vanished from one spot and appeared in another.

Pettini proposes something similar to explain entanglement. The particles aren't connected by a signal racing through space. Instead, the information linking them extends through a hidden second time dimension. From inside our own time dimension, the connection appears instantaneous.

There is an obvious problem. If ordinary matter could move through this extra time, bronze lions really might start teleporting. To prevent that, Pettini borrows a mathematical device from string theory known as a warp factor, which can stretch or compress dimensions. He arranges things so that the particles and forces of our standard model of

particle physics remain trapped in the four-dimensional world we know, while information associated with entanglement can extend into the hidden temporal direction.

It is fair to say this is all bold stuff. Yet, crucially, Pettini says the idea can be tested. In a paper from June that hasn't yet been peer-reviewed, he proposes using two independent pairs of entangled particles of light, known as photons. Photons in each pair would be measured by detectors that are placed far apart. Under standard quantum mechanics, the two pairs of photons have nothing to do with one another, so the results from one measurement experiment should bear no relation to those from the other.

His model predicts otherwise. In Pettini's picture, all entanglement reaches through the same second time dimension. That means two photon pairs created entirely independently could still interfere with each other's measurements, producing a tiny amount of "crosstalk" – and a small correlation between the results of both detectors. "This is a falsifiable statement," says Pettini.

Sören Petrat, a mathematical physicist at Constructor University in Bremen, Germany, welcomes the attempt. "It's a bit of an exotic suggestion," he says. "But from a scientific perspective, it looks like he has done everything right: suggest a solution to a deep problem, discuss its consequences and propose experimental tests."

If a correlation appeared in such a test – and that's a big if – it would be tempting to see this second time dimension as a potential new route through reality. Pettini has already allowed himself to consider the outrageous possibility that we could somehow harness it. "For fun", he is applying the idea to a possible warp drive, a hypothetical device that would move faster than light by distorting space-time around it.

For now, though, none of this means that a second dimension of time really exists. It may ultimately prove to be no more than a useful mathematical device – a way of revealing connections that are difficult to see in the four-dimensional world we experience. But what once looked impossible may one day explain physics' remaining mysteries. ■



Miriam Frankel is a journalist based in London

Can we ever end animal testing?

Better ways of screening drugs could soon lead to thousands fewer animal tests, finds **David Adam**

AS MANY as 190 million animals are used in biomedical research around the world every year. But after centuries of experiments, controversy and countless life-saving benefits, science may finally be approaching a point where animal testing is no longer the default option.

Until recently, that idea would have sounded implausible. Modern biomedical research has relied on animals almost since its inception, using them to study disease and test new medicines before they ever reach people. But over the past few years, that assumption has begun to shift.

Efforts to replace animals in science have the firmest backing yet from policy-makers and regulators. In April 2025, the US Food and Drug Administration (FDA) published a five-year road map to reduce animal testing in drug-safety studies. Months later, the country's National Institutes of Health (NIH) stopped requiring animal models for many of its grants. Late last year, the UK launched a similar plan to phase out animal testing, and the European Union followed suit in June.

These initiatives largely seek to make more widespread use of better alternative methods. A sliver of plastic no bigger than a microscope slide threaded with human liver cells can now identify toxic drug effects better than using rats. Lab-grown cells are sparing rabbits from once-indispensable tests that screen medicines for contaminants. And AI models are identifying potential drug targets at record speeds, without using animals.

"The science has reached a point where there is a much better evidence base to support the fact that phasing out much animal testing might be possible," says Celean Camp, head of the UK campaign group Replacing Animal Research.

Animal experimentation dates at least as far back as the ancient Greek philosopher Aristotle, who dissected dozens of animal species to better understand anatomy. But modern systematic use of animal experiments didn't start until the mid-1800s. Then, and now, advocates for using animals argued that many of them have bodies similar enough to humans to justify tests on biological mechanisms and drug effects, thus informing science and improving healthcare without injuring people.

Pushback started almost immediately. Public horror at some of the Victorian physiologists' methods prompted the UK to pass the world's first animal testing law in 1876, which ensured it was done only by licensed scientists.

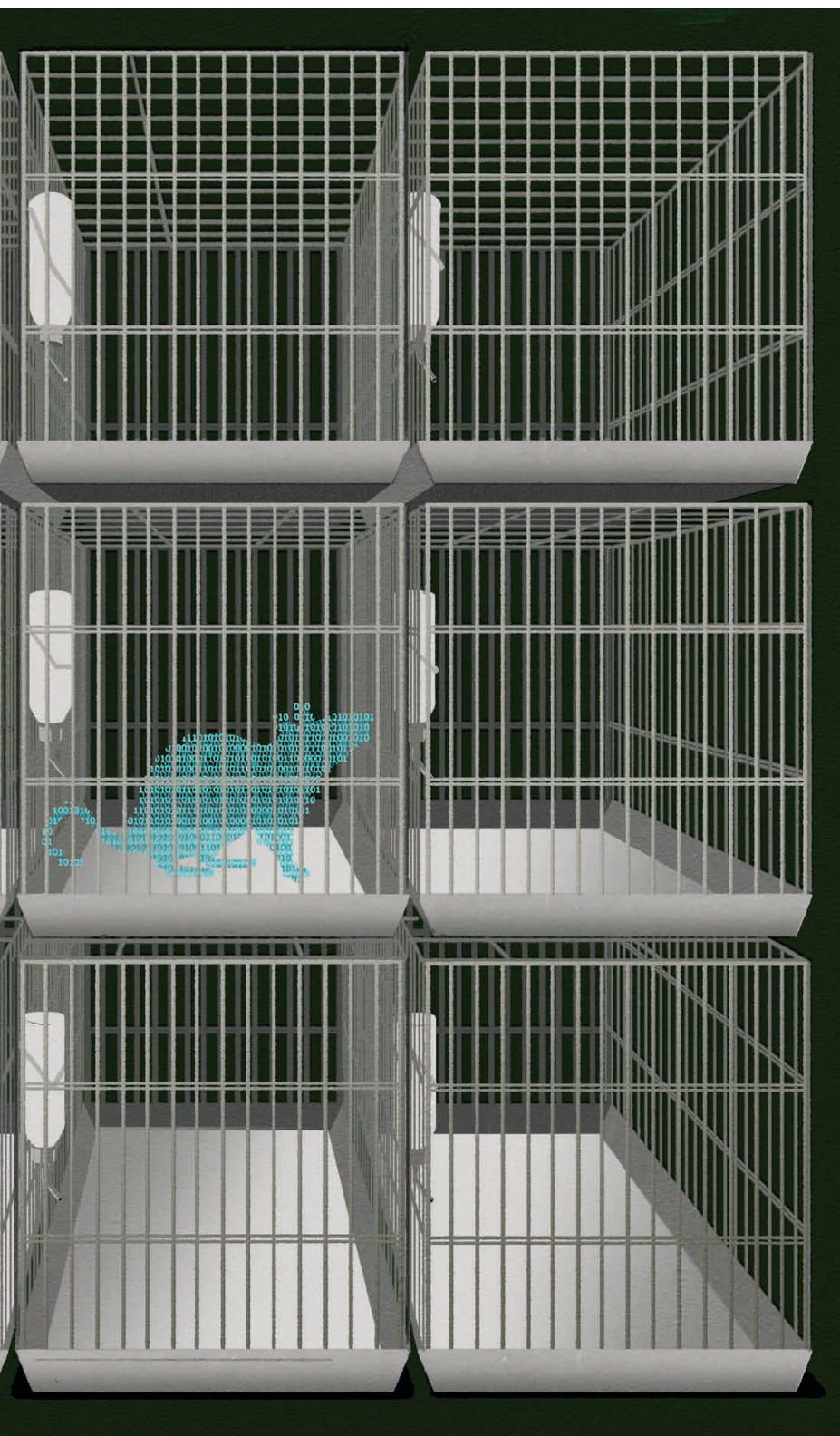
Yet, as generations of scientists have repeatedly said, nobody really wants to run animal experiments. They are slow, expensive and come with significant paperwork – and that is before everyone involved weighs up the ethics of deliberately inflicting pain on animals, including dogs, pigs, monkeys and, yes, guinea pigs.

But, as distasteful as it might be to some, animal testing has let researchers probe biological processes and screen experimental medicines. It has facilitated monumental breakthroughs, from the discovery of insulin and antibiotics to the development of pacemakers and covid-19 vaccines.

Most modern regulations on animal testing rest on a 1959 concept known as the 3Rs: replacement, reduction and refinement. "Replacement" means replacing animals with computer models or bacteria where possible; "reduction" is using as few animals as possible; and "refinement" is minimising pain and distress. Countries often vary in how they apply the framework. Rules in the UK and the EU, for



PETE RYAN



example, cover all vertebrates, but the US excludes mice, rats and birds bred for research, which constitute the majority of test animals.

But there are still a lot of animals used in research (see “How many animals are used in scientific research?”, page 34). About half of animals used in scientific tests these days are involved in basic research – the kind done at universities to uncover how biology functions and fails. The other half is split roughly between testing new drugs for side effects and setting safe exposure limits for chemicals, such as pesticides.

With some exceptions, public and political opinion has largely tolerated that trade-off. But there are signs of waning support. In 2001, most Americans believed medical testing on animals was morally acceptable. By 2025, fewer than half did. The most recent UK survey, from 2018, found support holding steadier at about two-thirds – but also a growing appetite for alternatives, especially among younger people.

“The public has always been somewhat supportive of animal research where there is no alternative,” says Camp. But new techniques now lessen the need for an increasing number of animal tests, and scientists haven’t caught up with that, she says. “There’s more and more evidence that the regulatory mechanisms to check that there is no alternative have not been functioning as well as they could,” says Camp. “So that assurance on which the public trust and the public support stands is a little shaky.”

Regulatory agencies are now trying to accelerate the transition to non-animal methods – and have been for a while. “The whole push towards more non-animal models isn’t recent,” says Ivan Rusyn, a toxicologist at Texas A&M University. Instead, he says, the recent proposals result from a “perfect storm” of political, scientific and regulatory activity over more than a decade. For example, back in 2013 the EU banned animal testing on cosmetics, proving regulators could force the issue and that industry would adapt.

The recent UK and EU road maps aim to build on this progress and develop more non-animal regulatory tests. The US approach is slightly different and aims to replace animals in scenarios where they are notoriously poor models – such as for testing monoclonal antibodies, a treatment that mimics the immune system.

Neither transition intends to be a wholesale replacement, as none of the alternatives can fully recreate a whole-bodied animal. But used in the right context – replacing an ➤

How many animals are used in scientific research?

These figures are for the year 2015 in the UK. In the US and Australia, comprehensive statistics are not reported.

2.5m

Total number of procedures

Some 48 per cent of these procedures were to breed animals for research, and about 52 per cent involved experiments on animals.

Mice

1,775,506



Fish

381,443



Birds

130,510



Rats

126,500



Some animals have special protection under UK law and were used in much smaller numbers:

Horses

12,101

Dogs

2889

Monkeys

2236

Cats

91

SOURCE: UNDERSTANDING ANIMAL RESEARCH

experimental method here or a safety check there – the technologies can meaningfully reduce animal testing and even produce better results.

So, what actually are the alternatives? One of the most established is cell cultures, which let researchers expose living human tissue to chemicals. These can replace some mandated drug and chemical toxicity tests, but a flat dish of cells is a poor substitute for the changing environment of a working organ.

Devices called organs-on-a-chip try to close that gap by recreating some of that dynamism. Roughly the size of a USB stick, they house cells on a flexible scaffold lined with tiny fluid-filled channels that supply nutrients and remove waste. Organs-on-a-chip have a significant leg up on animal models when it comes to testing drug safety. More than 90 per cent of experimental drugs that appear safe and effective in animals don't receive regulatory approval, largely because they turn out to be toxic to humans.

In 2022, biologist Donald Ingber at Harvard University and his colleagues at the biotech company Emulate, which he co-founded, built and tested the toxicity of experimental drugs using a liver-on-a-chip. It identified nearly 7 out of every 8 drugs that proved toxic in clinical trials despite appearing safe in animal tests. "People are ignorant about how far this field has progressed," says Ingber.

The chip, which is now in the final stages of FDA approval, has already made its way into labs. The pharmaceutical company Moderna has used it in place of primates to screen compounds used in mRNA vaccines for liver damage.

Ingber's lab also builds chips to mimic other organs, including the cervix, lungs, kidneys and intestines, and wires them together with artificial blood vessels. "We have, like, a human body-on-chips," he says. Mounted on a scaffold, they can physically flex to mimic breathing or gut motion, addressing one of the oldest complaints about cell cultures: that real tissues never work in isolation.

Even more complex are organoids, self-assembling blobs of human cells that mimic organs. Rather than growing cells on an engineered scaffold, as chip-builders do, organoid researchers stimulate stem cells – which can transform into other cell types – into miniature, three-dimensional tissues. The resulting "mini organs" capture more of a tissue's structure and cell variety.

Hans Clevers at Utrecht University in the Netherlands, a molecular geneticist who

pioneered organoids, argues the technology can already outperform animal models for some human conditions, including cystic fibrosis. "We have animals that beautifully mimic the symptoms, but they typically don't have the underlying cause," he says.

That distinction matters because laboratory animals have always represented a compromise. Scientists can coax mice to develop cancers, inflammatory diseases and neurological disorders resembling human illness, but such conditions often arise through genetic engineering or artificial intervention, rather than the biological processes that make people sick. Clevers says his team developed an antibody to treat a head and neck cancer called squamous cell carcinoma entirely in organoids, with no conventional animal testing. The drug is now in phase III trials.

Organoids and organs-on-a-chip also open a door to personalised medicine in a way that animals can't. Researchers can grow them using a patient's own cells and then test candidate drugs directly on tissue sharing that patient's biology.

Digital simulations

Other animal alternatives don't involve biological material at all; they run on computers in what researchers call in silico models. AI systems trained on genetic, chemical and clinical datasets can flag potential drug targets and predict toxicity. One problem, though, is that models don't always explain how they generate decisions, which makes regulators cautious about relying on them to judge safety.

Molecular docking software goes further and simulates how a potential drug physically fits and binds to its target protein, letting chemists screen thousands of compounds on a laptop. Then there are pharmacokinetic models, which divide the body into mathematical compartments – liver, kidneys, bloodstream – to predict how they absorb, break down and clear drugs. The approach could help estimate safe doses in groups of people rarely included in trials, such as children or pregnant women.

None of these tools alone gives the complete picture of a drug. But together, they can help researchers zero in on compounds most likely to succeed and then investigate them further with organoids, organs-on-a-chip and remaining animal tests.

In practice, it can still take years, or even decades, for non-animal methods to be widely adopted. Take fever tests, a widely used way of



checking whether drugs contain contaminants. For more than a century, regulators have mandated injecting new drugs into rabbits to see if they cause the animals to develop a high temperature, signalling an infection – simple, effective and unpleasant for the rabbit.

Alternatives to using rabbits have existed for decades. For example, a screening test using horseshoe crab blood, which clots on contact with bacterial toxins, dates to the 1970s. Then, a non-animal test emerged in the 1990s: a human blood test that measures

immune responses to contaminants.

Yet, it took until this year for the rabbit test to be officially phased out, and then still only in Europe – it is ongoing in some countries, including the US.

Some of the lag comes down to understandable caution. “The regulators say they can only make judgements on whether the data is adequate once people submit it to them,” says Camp. “And the companies are saying, well, we’re only going to submit data that we know the regulators are going to accept.”

The new road maps are attempting to break this impasse by validating non-animal models and creating guidelines around their use in drug development.

It can also be challenging for career scientists to move from familiar territory. “Academics generally tend to do incremental stuff, and it’s very hard to learn a whole new methodology,” says Ingber.

There are some signs of movement, however. A 2024 analysis of research on breast cancer, lung disease, blood cancer, heart disease, neurodegenerative disease, diabetes and toxicology found that by 2022, publications using only non-animal methods had overtaken those using animals in all seven areas. That could be due to a rise in tests developed as a response to the EU cosmetic-testing ban.

Government money may also accelerate the shift. China launched a project in 2024 to speed up the development of lab-grown human tissue models. In the US in 2025, the

NIH opened its first organoid-development centre, while halting funding for research using only animal models. In Australia, the New South Wales government allocated \$4.5 million in 2024 for a programme to reduce and replace animals in medical research.

None of this is to say that animal research will ever be phased out entirely. “It’s animal research that gives us an understanding at the organism level that all of this in-vitro stuff, including organoids, do not,” says molecular and cell biologist Michael Hadjiargyrou at the New York Institute of Technology. Delete a gene from cells in a dish, and the consequences are often minor. Delete it from a living animal, and it can reshape their physiology.

“We need to educate Congress. They need to know how critical animal research is to scientific discovery,” says Hadjiargyrou.

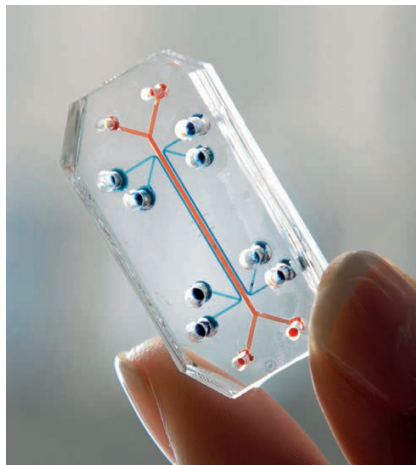
Rusyn thinks political rhetoric on reducing animal testing has occasionally run further than the evidence justifies. He is particularly unimpressed by a US Environmental Protection Agency’s pledge to drop mammal testing requirements for pesticides by 2035. “It is completely bonkers, in my opinion, as a toxicologist,” he says.

Because cell cultures are often more sensitive to chemicals than an intact animal, they risk banning substances that would otherwise be harmless at the doses people encounter. Even committed opponents like Camp accept that some animal research will always be necessary. It is difficult to see how research into psychiatric and neurodegenerative diseases, for example, could proceed otherwise, given the complexity of the central nervous system and the brain.

But there is some evidence that the regulatory drive for replacements is having a positive impact. In its first-year progress report, the FDA announced that a change to primate toxicity testing for certain immune-based therapies had saved hundreds of non-human primates and that there is “potential for [saving] thousands more animals as methods expand”.

But while the shift to alternative methods may be a long and laborious process, there are at least 190 million reasons to keep trudging ahead. ■

Devices like this lung-on-a-chip get accurate results for drug testing



WYSS INSTITUTE AT HARVARD UNIVERSITY



David Adam is an author and journalist who covers science, technology and medicine



PHOTOGRAPH BY RICHARD EYRE

The wellness boom has brought a wide array of consumer products that promise to boost your health and happiness. Doctor and presenter **Xand van Tulleken** tells Grace Wade the three trends to be most wary of, and why

IN 1893, a cowboy named Clark Stanley killed hundreds of writhing rattlesnakes at the Chicago World's Fair. He then tossed them into a boiling vat and scooped up the rendered fat to sell as his now-infamous snake oil, which he sold as a tonic for all manner of health problems.

This particular scam wouldn't work today, but discerning quackery from legitimate medicine has become, if anything, even more of a challenge. To add to the confusion, healthcare advice often has to compete with the vaguely defined wellness industry, which, thanks to the rise of influencers and social media, is highly persuasive.

Doctor and presenter Xand van Tulleken looks into this unregulated industry in his latest book *Make Me Well: A quest to understand the confusing world of wellness*. He spoke to *New Scientist* about his findings, revealing what does – and doesn't – keep us healthy.

Grace Wade: What exactly is "wellness"?

Xand van Tulleken: Wellness is about the way

"The modern wellness industry really tries blurring the boundary between health and wellness"

we feel. It is about increasing our performance, optimising our bodies and, to a degree, reducing our risk of disease. But wellness practices are not healthcare.

What is the distinction? I always thought of wellness as optimising health.

The modern wellness industry really tries blurring the boundary between health and wellness. It is the reason I wrote *Make Me Well*, because I was struggling to distinguish between the two.

Health is typically about a defined problem that is measurable, and healthcare aims to prevent and treat specific conditions. Wellness, on the other hand, tends to work on things that are more slippery, that are harder to pin down. It is about feeling your best, optimising your performance, looking better.

The wellness industry isn't allowed to claim its products can treat medical problems, with the exception of some tightly regulated scenarios for certain supplements, such as that vitamin C contributes to the reduction

of tiredness and fatigue. But all the restrictions on advertising and marketing and labelling don't restrict influencers from giving their opinions.

Another way to think about it is that healthcare is interested in the floor, while wellness is interested in the ceiling. Healthcare wants to stop you from falling below a certain level. Its aim is to keep you within normal range. Wellness dangles your best self out in front of you. It asks you to imagine the ceiling of how you can feel.

Are there any wellness products that improve health? Becoming the best version of myself certainly sounds appealing.

I focused on three areas of wellness: supplements, wearable technology and the testing industry. The hallmark of these products is that they look like medicine, but they aren't. For almost everything, when I dug into the evidence, it evaporated.

None of these products are intended to treat, diagnose or improve any disease or

health condition. There is no testing for efficacy or safety. The wellness industry doesn't have to prove efficacy, because it doesn't claim these products do anything.

Of course, it is very clever at making health claims through routes that aren't legally scrutinised. You can't put a health claim on a vitamin bottle, but you can get a podcaster to say it.

Is there any harm in trying these products, like vitamins, for example?

Yes, you need vitamins. But you don't need vitamin supplements. That is one thing I feel strongly about: vitamin supplements don't work for healthy people. That was hard for me to come to grips with, because it went against my intuition.

Doctors, including myself, were on board with supplements for a long time. We assumed they were completely safe and that your body could manage excess fairly well. But then these big trials in the 2010s revealed that multivitamins aren't associated with a ➤

reduction in death from any major health condition. And specific antioxidant supplements can actually increase mortality by roughly 4 to 16 per cent, depending on the exact vitamin and dosage.

Antioxidants seem to increase cancer risk by shielding cancer from destruction, but I don't know why large quantities of fruits and vegetables don't pose the same risk. For some reason, our bodies seem to process the nutrients quite differently.

Now, vitamins are a fantastic way of addressing specific health problems, and doctors use them to treat different illnesses. You have beriberi? That is a vitamin B deficiency. You have rickets? That is commonly vitamin D deficiency.

But if you have a vitamin deficiency, you need to see your doctor. That is not a place where the wellness industry should be interfering. If you are low in iron, your doctor won't just prescribe you a supplement. They will also ask why. Do you need a colonoscopy? Do you have pernicious anaemia? Is your menstrual cycle irregular? There are different ways to think about the problem that a doctor is incentivised to look for that a company selling supplements isn't.

What about at-home tests? It seems like it would be useful, not harmful, to screen for deficiencies and illnesses.

The testing industry sells tests for cancer, menopause, hormones, vitamin deficiencies and so forth. Yet there is good evidence that blindly screening for disease doesn't save lives and can actually make people less healthy.

As an example, there was a large trial of a blood test for prostate cancer in Finland. Only about a third of those who were positive had cancer upon biopsy. That is why you only get tested when there are other reasons to believe you have prostate cancer. Otherwise, you risk further investigations that can be harmful, that can cause nerve damage, that can create unnecessary anxiety.

When your doctor tests you for something, they will have worked out the likelihood of you having a certain health condition beforehand. Context matters.

"I am not anti-wellness. I just want to save people time and money and set them free to do what matters to them"

NewScientist Live



Hear more from Xand van Tulleken at this year's New Scientist Live.

On 11 October, he and his brother Chris will delve into the wellness industry, including tips on how to tell the good advice from the bad.

For more information, visit newscientist.com/nsmlag



RAGHU RAJAMAGNUMPHOTOS

What about wearable technology? I have my smartwatch right here, and while I'm sceptical of how useful all this data is on my heart rate, stress levels and sleep, I don't think it has harmed me.

I think the appeal of those numbers is that they condense the terrible complexity of life into a simple, magical number. When I am on my exercise bike, it is the only moment in the day where I know exactly how I am doing. I know my heart rate. I know how many watts I am putting out. I have a totally honest account, whereas, right now, I don't know how I am doing in this interview.

I think you are doing a good job.

That is good – yet I want an objective score out of 10!

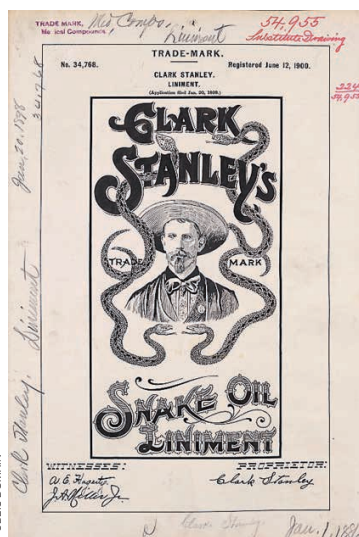
But to get to your point, I think there are two different ways to think about wearable technology. Although they can create an unhealthy relationship with data – maybe people chase their step count or develop an exercise dependency – for most people, they just give you a number. There is no evidence they will make you fitter.

But I am a big fan of democratising health information. I like knowing about my body. I think it is similar to owning a set of bathroom



**Doing something
you love – like
martial arts – is good
for your wellbeing**

**Below: The original
snake oil was created
by a cowboy called
Clark Stanley**



scales. You probably know which way the dials should be trending.

It gets sinister and dangerous, in my opinion, when these devices screen for illness. Many wearable technologies are approved [by the US Food and Drug Administration as] medical devices, but the medical industry puts them to very different use than the wellness industry.

Wearing a glucose monitor if you have diabetes can save your life. Wearing a glucose monitor for “wellness” could actually harm you, because you then try to flatten or remove blood sugar spikes that are normal, physiological responses. It generates alarm about foods that aren’t harming you, leading you towards eating less fruit, fewer legumes, fewer whole grains, more cheese and meat, which makes your cardiovascular risk worse.

I once bought two simultaneous subscriptions to the same personalised nutrition app. I wore two glucose monitors, sent two scoops of the exact same poo, drew blood from each arm on the same day. I sent in the same exact data, yet got completely different results on my blood sugar and microbiome and completely different advice about what I should eat. So you are getting nonsense back.

If the wellness industry is full of nonsense, why do you think interest in wellness has exploded in recent years?

I lay a lot of the blame at the feet of the medical industry. I do think the medical profession has many failings that have pushed people toward wellness. It is difficult for a lot of people to access healthcare. Even in places with universal healthcare, such as the UK, it is hard to get an appointment, and when you do, you only get so much time because there are not enough doctors.

Part of it, too, is the medical industry lets certain groups of people down, particularly women and minority groups. So even if they can access healthcare, they have a difficult time getting the treatment they need. Look at menopause: consumers spend billions of dollars each year on menopause supplements. Do you know how much the US spends every year on menopause research?

I want to guess \$100 million, but that feels too small.

We would dream of \$100 million! Instead, it spends \$57 million a year. We don’t even record how much we spend in the UK, because we basically spend nothing at all. So there is no money for this.

Now, imagine going through menopause, which can be extremely difficult, and the scientific establishment just isn’t interested in that particular problem. It is no surprise, then, that you would turn to the wellness industry. Many people’s experience with healthcare is that it is disempowering and paternalistic, and I think people want to feel empowered.

Doctors are also constrained in what they can offer. When you buy a supplement, there is an aura of possibility it will make you live forever. When you buy a drug, your doctor says: “There is a 20 per cent chance it will reduce your risk of heart attack in the next 10 years by 5 to 7 per cent. Also, here is an endless list of potential side effects.” It isn’t exciting.

What advice do you have for people wanting to improve their wellness? Or are we tough out of luck?

Make use of healthcare professionals before you buy a single wellness product. Have you had your colonoscopy? Do you know your blood pressure? Are you eating your vegetables? These straightforward basics are what improves wellbeing.

You should also think of wellness beyond health. It is a way of feeling good that should be grounded in community and family and friendships and personal growth. I am not anti-wellness. I just want to save people time and money and set them free to do what matters to them.

Find something you love and pursue it with enthusiasm and passion. That is the best bit of wellness advice I have come across. Whatever it is – martial arts, knitting, dancing, pottery – maybe that is the wellness metric you want to aim for. And if biohacking is your passion, go for it. That is up to you. But if you are casting around wondering whether some blue zone, supplementary, special-treatment, stem-cell stuff is going to extend your life, I wouldn’t bother. I would disregard all of that. ■



Grace Wade is a features editor focusing on health and biomedical sciences

Ever-more distant cousins

As we find older and older fossils of ancient hominins, the timing of our split from chimps is being pushed further and further back, finds **Michael Marshall**

CHIMPANZEES, along with bonobos, are our closest living relatives. Of all the millions of other animals on Earth, they are the ones that are most like us. And the more we learn about chimpanzee behaviour, the more obvious the similarities become.

For instance, wild chimpanzees peer at each other: one chimpanzee will sit close to another and closely watch what they are doing. Nora Slania at the Max Planck Institute of Animal Behavior in Germany and her colleagues found that, from a young age, chimps peer at others as they perform a range of tasks. This included complex tasks like making tools, but also supposedly simple things like feeding and grooming.

The implication is that a great deal of chimpanzee behaviour is culturally learned by watching others. Obviously, chimps will instinctively try to eat food, but it isn't obvious what is good to eat, or how to peel a fruit or crack a nut. One way to learn is by watching how the adults do it.

Some chimpanzees have figured out how to eat army ants, which form huge colonies comprising millions of individuals. When the ants need to eat, they assemble into marching columns guarded by soldier ants with powerful jaws. While there is obviously plenty of nutrition in an army ant colony, one must be careful when dealing with them.

In Senegal, chimps catch *Dorylus* army ants living on the savannahs. Andreu Sánchez-Megías at the University of Barcelona in Spain

and his colleagues argue that chimps vary their strategies depending on how aggressive the ants are. Army ants that hunt in leaf litter aren't very aggressive, so chimps sometimes forage for them by hand. In contrast, army ants that hunt on the surface of the ground are aggressive, so the chimps catch them by lowering a stick, which the ants bite onto – allowing the chimp to eat them in relative safety. What's more, the chimps prefer to catch these species while standing upright or on a raised platform, like a bent tree branch. These are all savvy ways to minimise your chances of being bitten.

One more example: centuries ago, chimpanzees in Sierra Leone may have used tools to forage for oysters. They don't do so today, because chimps no longer live on the Sierra Leone river estuary, so there is nowhere for them to do it. But reports from 1678 and 1682 describe apes – whose description best matches chimps – using sticks to dislodge large oysters. If this is true, it adds to the evidence that chimps sometimes go, for want of a better word, fishing. Chimpanzees in the Nimba mountains of Guinea have similarly been seen fishing for crabs. Other populations have developed a host of methods for gathering algae.

These behaviours are creative and flexible, requiring careful observation and extrapolation – all traits we associate with ourselves. They are a reminder of how close chimpanzees are to us. So it is curious that,

in parallel, genetic and fossil evidence has gradually pushed chimpanzees further away from humans.

I'm talking about the last common ancestor of humans and our two closest living relatives, chimpanzees and bonobos. We know this ancestral animal was an ape, and we know it lived millions of years ago, but we don't know what it was like or exactly where and when it lived.

In a timeline of human evolution I recently compiled for *New Scientist*, I gave a wide date range for the last common ancestor – and for many other key events in prehistory. That is because many of these things can't yet be precisely pinned down. Often, different lines of evidence – for instance, physical fossils or genetics – give different answers.

Hence, I put the last common human-chimpanzee ancestor sometime between 13 million and 5.5 million years ago. That is a range of 8.5 million years, which, in this context, is pretty wide. For comparison, the oldest putative fossil hominin is *Sahelanthropus* from about 7 million years ago, so the uncertainty I gave for the last common



IMAGEBROKER.COM GMBH & CO. KG/ALAMY



ancestor spans more time than the known hominin fossil record.

Somehow, an ancient population of apes became divided, with one group going on to become modern chimpanzees and bonobos, and the rest becoming humans and all other hominins. Here, I'd like to drill into exactly when and how this happened. I am basing this on a study published in the journal *Primates* on 18 June, which synthesises all the estimated dates for the split between humans and apes.

Lead author João d'Oliveira Coelho is a biological anthropologist at Gorongosa National Park in Mozambique. A few years ago, he was writing about his research and wanted to include a quick sentence stating when the hominin lineage began, so he looked around and found a smorgasbord of estimates – and some peculiar statements. Textbooks, even quite recent ones, often put the human-chimp split as recent as 4 million years ago, but would then mention the existence of significantly earlier hominins like *Ardipithecus*, *Orrorin* and *Sahelanthropus*. “How can you have both these statements in the same book? They are completely

Above: Chimpanzees teach each other to use tools to access food

Below: The oldest putative hominin fossil, *Sahelanthropus*, dates from around 7 million years ago



WORLD HISTORY ARCHIVE/ALAMY

contradictory,” says d'Oliveira Coelho.

He and his colleagues began systematically compiling estimates of the human-chimp split, aided by a handy online database called TimeTree, which compiles estimates of all such “species divergences”. Eventually, they had 202 estimates, published from the late 1960s to the 2020s. Nobody else has systematically analysed the estimates in this way, says d'Oliveira Coelho (and I can't recall seeing anything comparable).

At first glance, the uncertainty range hasn't narrowed, or at least not much. If you just look at estimates prior to 1980, they range from 4 million to 14 million years ago. Estimates published from 2015 onwards vary from 4 million to 12 million years ago. Superficially, not much has changed.

Our family tree

However, don't let the outliers fool you – on any question that attracts a lot of attention, there will always be some wild results. The important thing is the average.

D'Oliveira Coelho found a clear trend: the estimated timing of the human-chimp split has been steadily moving further back in time. Early studies had an average age of about 6 million years; more recent ones have pushed it back to around 8.5 million years. In other words, we may have been evolving separately from chimpanzees for 2.5 million years longer than we thought.

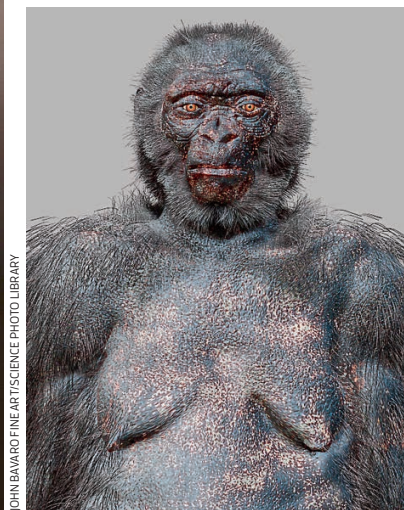
Why has the date moved so far back? Some of it is simply a matter of better data. We got the first draft of the chimpanzee genome in 2005. Everything before that was based on snippets of the genome, not the whole thing. Twenty years later, we got a much more complete version of the chimp genome, which filled in a lot of missing pieces and corrected many errors. Nowadays, we also have a great many human genomes, meaning that we have a better sense of how much the DNA sequence varies.

But it is also about the underlying methods. You can estimate how long two species have been evolving separately by comparing their genomes. If the DNA sequences of both species are very similar, they probably split only recently, and if there are lots of differences, they split a long time ago. In essence, it's simple.

In practice, it's fiendish. Different parts of the genome evolve at different speeds. ➤



The discovery of *Ardipithecus* fossils pushed back the timing of the human-chimp split



JOHN BAVARO/FINEARTS/SCIENCE PHOTO LIBRARY

"ONE TEAM PLACED THE SPLIT BETWEEN 8.69 MILLION AND 7.28 MILLION YEARS AGO"

Genes involved in immunity change at a furious rate to keep up with new diseases, while genes involved in fundamental processes like making proteins barely change at all. Some species' genomes change faster than others. Changes in population size matter. Changes in generation time – that is, the average age at which a species reproduces – also matter, and this can vary between males and females.

In response, geneticists have built complex models that include all these factors. The trouble is, figuring out just one of these things – say, the generation time – is a research project in itself. Inevitably, one has to make assumptions: 100,000 years ago, did *Homo sapiens* still have the same generation time as today? What about the ancestors of chimpanzees 3 million years ago?

"There's all these things you need to model, and they are not in straight lines," says d'Oliveira Coelho.

However, one of the biggest factors in pushing the date of the split back is the gradual

acceptance of key fossil finds. The oldest putative hominins have all been discovered in the 21st century. *Orrorin* and the oldest *Ardipithecus* were first described in 2001, with *Sahelanthropus* following in 2002. This pushed the fossil evidence of humanity way back, from around 4 million years ago to 7 million years ago.

In theory, this should have immediately pushed back genetic estimates. Geneticists can use well-dated fossils as "anchors" in their models: instead of guessing how long a particular branch of the family tree was, they can input a hard data point. But for a long time, they didn't.

"Even after the three oldest hominins, *Ardipithecus*, *Orrorin* and *Sahelanthropus*, were published, it took years and years and years for geneticists to include that," says d'Oliveira Coelho. *Sahelanthropus* is about 7 million years old, but it wasn't until around 2015 that the average estimate of the human-chimp split became older than this. D'Oliveira Coelho blames this on a lack of communication between physical anthropologists and geneticists, and on some geneticists being overconfident about their data and methods.

His team ran its own meta-analysis, which placed the split between 8.69 million and 7.28 million years ago. However, d'Oliveira Coelho emphasises that this won't be the last word. The only thing he is confident about is that claims of a really recent split can be safely ruled out, based on the fossil

evidence. There is still a window of millions of years when it could have happened.

To sum up, 50-odd years of research has slowly pushed the date of the last common ancestor further back in time. This means that chimpanzees and bonobos are more distantly related to us than we thought: many more generations separate humans today from the ancestor we shared with them.

Yet the more we learn about chimpanzees, the more they look like us. A study published in March showed that, when chimpanzees play together, they can quickly mimic the expressions their playmates make when they laugh. This may make for smoother social interactions and more enjoyable play. A June study, by a different group of researchers, found that chimpanzees and other great apes laugh in ways that are remarkably similar to us. This was especially true of laughter elicited by tickling. The researchers argued that "great apes have been laughing in a recognizable way to modern humans for at least 15 million years".

I rather like that idea. ■

To read Mike's timeline of human evolution, visit: bit.ly/4xhTbwi



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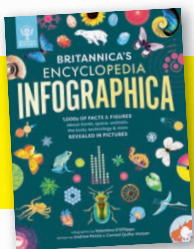
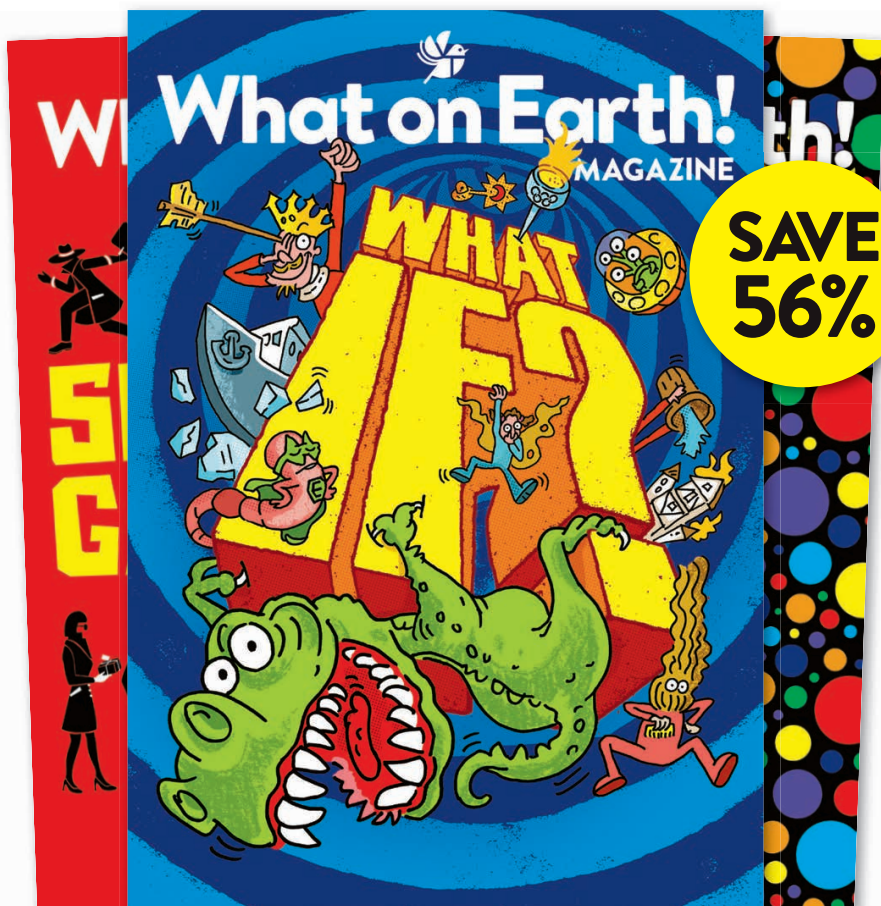
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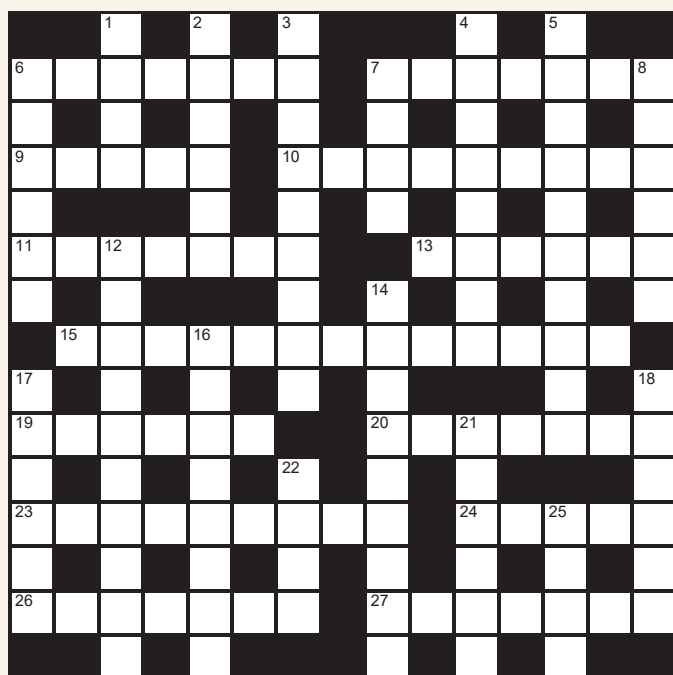
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Quick crossword #217 Set by Richard Smyth



Scribble zone

Cryptic crossword next week

ACROSS

- 6** Nitrogen-rich nutrient (7)
- 7** US tank (7)
- 9** Unit equivalent to just over 28 g (5)
- 10** Ear specialist (9)
- 11** Using or requiring air (7)
- 13** Hook-and-pile textile fastener (6)
- 15** Oviduct (9,4)
- 19** Cm (6)
- 20** Archaic form of chemistry (7)
- 23** The opposite of convexity (9)
- 24** ___ acid, 6-across component (5)
- 26** Reproduction type that does not involve the fusion of gametes (7)
- 27** Pruritus (7)

DOWN

- 1** Io, say (4)
- 2** Echoing effect, in music (6)
- 3** Kind of internal examination (9)
- 4** Reduce or remove tree cover (8)
- 5** Unable to form a homogeneous solution (10)
- 6** Fluoxetine brand name (6,6)
- 7** Form of urban air pollution (4)
- 8** Idea (6)
- 12** Ground cuckoo; nemesis of Wile E. Coyote (10)
- 14** Inability to move (9)
- 16** Tincture of opium (8)
- 17** Tree also known as a wattle (6)
- 18** Machine-augmented human, in science fiction (6)
- 21** Risk, probability (6)
- 22** Small item of laboratory glassware (4)
- 25** Pigmented ring-shaped structure in the eye (4)

Quick quiz #370

set by Chelsea Whyte

1 The LUX-ZEPLIN experiment in South Dakota may have seen the first hints of a dark matter particle. What particle might it have detected?

2 Which psychedelic compound has been shown to prevent a common side effect of chemotherapy?

3 A storm wider than Earth has been spotted at Saturn's south pole – what shape does it take?

4 Critically endangered pygmy raccoons on Cozumel Island in Mexico have shown remarkable creativity by making their own toy balls. What material do they use?

5 What explanation did a modelling study find for Venus having no moon?

Block Universe created by Puzzlr

The goal is to divide the grid into non-overlapping rectangles. Each rectangle must contain exactly one number, and have an area equal to that number. For example, a 6 should have a rectangle covering six squares.

				15				3
		9						
						5		4
			16		8			
								6
				5	2			
8				2				
					2		9	
5								

Answers to this week's puzzles on page 47



Our games are now playable online
[newscientist.com/games](https://www.newscientist.com/games)

Hot or cold

Why can't we cool water as quickly as we can heat it?

Hillary Shaw

Newport, Shropshire, UK

The reason is because our preferred temperature range is much nearer the freezing point of water than its boiling point. You can only raise or lower the temperature of something if you have something else that is hotter or cooler than the temperature you want to get to (second law of thermodynamics).

Room temperature is 15–25°C (59–77°F), our fridges are 0–5°C (32–41°F) and domestic freezers can be as low as -20°C (-4°F). A stove ring or kettle element can reach over 200°C (about 400°F), so we have no trouble getting water up to 100°C (212°F) given there is a temperature gradient of well over 100°C. But to cool water without freezing it, we only have the fridge, a temperature gradient of about 20°C.

Also, an electric stove ring uses 1–2 kilowatts, whereas a fridge uses just 5 per cent of that, so heat removal via a fridge is slower than heat addition via a stove ring. You could put your glass of water in a large freezer, but the temperature

“Much of the impression that cooling is slower is because we heat water actively, but often let it cool passively”

gradient is still just 40°C, much less than that provided by a stove. You also risk freezing the water, possibly cracking the container.

If terrestrial temperatures were 80°C (176°F), you could cool water (in a high-power fridge) quicker than you could heat it (before it boiled away).

Mike Follows

Sutton Coldfield, West Midlands, UK
It has certainly felt that way during the recurring heatwaves many people in the northern hemisphere



CREATE IMAGE/GETTY IMAGES

This week's new questions

Rock the boat If every ship on the planet were hauled out of the water, how much would global sea levels change, and how quickly? *Stacey Gambrill, Westgate-on-Sea, Kent, UK*

No O₂ life How different would Earth be if anaerobic life were the only life form, and could it have become sentient? *Dom Castner, Bellingham, Washington, US*

have experienced this summer.

When we heat things, we usually do so actively. A kettle can dump a couple of kilowatts of power straight into a litre of water. Cooling is often more passive: put a warm object somewhere colder and thermal energy leaks into the surroundings and, as the temperature difference diminishes, that heat flow slows. A fridge speeds up cooling by using energy to pump heat from the cold interior into the warmer room.

However, there is no fundamental rule that says water must cool more slowly than it heats. So, during our everyday uses of water, much of the impression that cooling is slower is simply because we heat it actively and quickly, but often leave it to cool passively and slowly.

The more interesting asymmetry appears when we try to cool something below ambient temperature. Then we encounter a less familiar problem: entropy.

Entropy is often described rather vaguely as “disorder”. Perhaps my fellow physicists will forgive me a little simplification here: for our purposes, think of entropy as baggage associated with transferring heat. If we want to cool something, that baggage has to go somewhere.

This is where the numbers become rather unforgiving. For a given amount of heat transferred, the associated entropy change is greater at lower temperatures. Transferring 1 joule of heat reversibly at 30 kelvin (-243°C) involves 10 times the entropy change of transferring 1 joule

How much would sea levels change if we removed all boats from the water?

at 300 K (27°C); at 3 K (-270°C), it is 100 times as much entropy change. At very low temperatures, even removing a tiny amount of energy therefore means finding a home for a relatively large amount of entropy.

Entropy, unfortunately, can't simply be thrown away. Once we want to cool something below the temperature of its surroundings, we have to move that entropy from somewhere cold to somewhere warmer, and that takes work. The colder things get, the harder that task becomes.

My old professor, the late George Pickett, advised thinking about temperature in orders of magnitude. Room temperature is about 300 K, while the centre of the sun is around 15 million K. On a logarithmic temperature scale, a factor of 10 is always the same interval on a graph: from 15 million K to 1.5 million K, or from 3 K to 0.3 K. But the physics of achieving those drops is anything but the same. As we approach absolute zero, removing thermal energy and disposing of its entropy becomes progressively harder, demanding increasingly ingenious cooling techniques.

I experienced that difficulty first-hand as part of Pickett's ultra-low temperature physics group at Lancaster University, UK, when we achieved the lowest temperature ever recorded in the known universe – an achievement rather undersold, I always thought, by its billing as a Guinness World Record.

Smelly illusions

There are optical illusions and audio illusions. Is there such a thing as an olfactory illusion?

Philip Martin

Tintwistle, Derbyshire, UK

I can personally testify that there is. After contracting the omicron variant of the covid-19 coronavirus, I lost my sense of



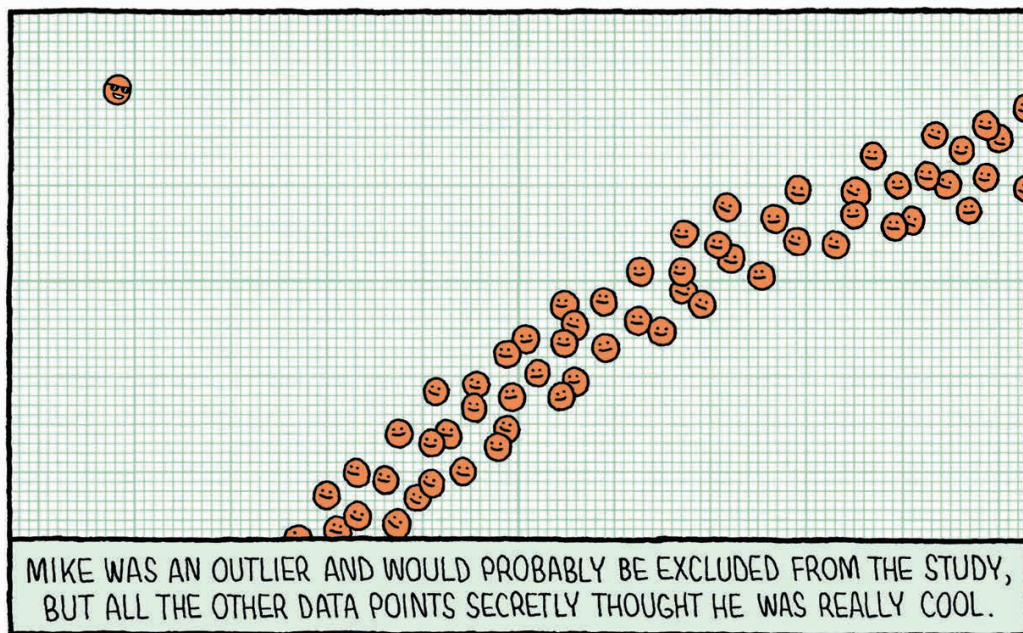
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Tom Gauld
for *New Scientist*



smell almost completely.

After several years hoping in vain that I would recover naturally, I was advised by my family doctor to try “smell retraining therapy”, developed by Thomas Hummel at the University of Dresden in Germany. This involves trying to teach your brain to relearn the range of core smell types: menthol, citrus, rose, etc.

Although initially dubious, since consistently inhaling essential oils over the past few months, I have once again begun to detect many odours, such as perfume and various foodstuffs. However, during this period, I have also experienced some episodes where strong smells have appeared randomly out of nowhere with no obvious link to anything in the surrounding environment.

Interestingly, at the same time, I experience intense memories of places associated with those smells. It seems as if, during the process of relearning, my brain is mixing up sensations and generating phantom smells.

“I have experienced episodes where strong smells have appeared randomly with no obvious link to the surroundings”

Joe Oldaker

Nuneaton, Warwickshire, UK

Almost certainly. I heard of someone who constantly smelled the scent of daffodils, although there were no daffodils present. It turned out he had a brain tumour.

Hear that?

Why don't dogs look up when a low-flying plane or helicopter passes overhead? I've never seen one do this. (continued)

Guy Cox

Sydney, Australia

Most previous replies have been about individual dogs. I'm afraid this is another, but it may have a more general application.

My daughter's dog Guffy, a kelpie/spaniel cross, is terrified

of thunder. If he hears distant thunder, he runs and hides. However, he also mistakes the rumbling of a plane going over for thunder and does the same – with no thought of looking up.

There seems to be some logic to a dog fearing thunder, as it can indicate a coming drenching and loud thunderclaps that must be painful to a dog's sensitive ears. Dogs have lived with thunder for much longer than they have lived with planes.

Harry Olund

Newport, Oregon, US

I have had four Labradors in the past 60 or so years. They all looked up when a plane flew overhead – if only for 4 or 5 seconds. It didn't hold their interest for very long.

Ron Bennett

Yorkshire, UK

Reading about West Yorkshire dog Blu, who tracked low-flying planes (1 August), makes me think there must be something in the area's air. We had a rescue collie that used to chase planes. ■

Answers

DOWN 1 Moon, 2 Reverb, 3 Endoscopy, 4 Deforest, 5 Immiscible, 6 Prozac, 7 Smog, 8 Notion, 12 Roadrunner, 14 Paralysis, 16 Laudanum, 17 Acacia, 18 Cyborg, 21 Chance, 22 Vial, 25 Its

ACROSS 6 Protein, 7 Sherman, 9 Ounce, 10 Otologist, 11 Aerobic, 13 Velcro, 15 Fallopian tube, 19 Curium, 20 Alchemy, 23 Concaivity, 24 Amino, 26 Asexual, 27 Itching

#217 Quick crossword

1 A WIMP
2 Psilocybin
3 A decagon
4 Rubbish
5 The planet consumed it

Quick quiz #370

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			2	5	
6					
			8	16	
4			5		
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Block Universe Solution

Paper writer

If you are a researcher studying a niche topic, one of the few ways to get some attention is to give your paper a witty title featuring a pop culture reference. Feedback has long been a fan of such things, but we would never have thought to do what the team led by Gabriel Budel at Delft University of Technology in the Netherlands did, which is to get systematic about it.

Budel and his colleagues searched two academic databases for papers whose titles contain references to the titles of songs by the Beatles. They ended up with "2237 exact references and 963 instances of creative wordplay".

That is a lot of Beatles jokes, so without further ado: the most-used Beatles song in academic paper titles is *The Long and Winding Road*. Budel's team found 798 examples of this, such as "Do black fintechs matter? The long and winding road to develop inclusive algorithms for social justice". It was followed by *With a Little Help From My Friends* (209), *Here, There and Everywhere* (129) and *Here Comes the Sun* (103).

Feedback feels that many of these phrases are quite easy to incorporate. We are more impressed by the 15 research groups apiece that managed to use "Glass onion", "Strawberry fields forever" and "Norwegian wood (This bird has flown)". The last of these gave rise to the frankly majestic "Norwegian wood, isn't it good? Narratives of the lumber industry and development paths in the Nordic periphery". We thought it would be really tricky to use something like "Everybody's got something to hide except me and my monkey", but then we checked and found three papers that explicitly play on this title.

The authors also performed a follow-up study looking for paper titles that use Beatles lyrics, yielding such gems as "The hyphen as a segmentation cue in triconstituent compound processing: It's getting better all the time" and "When I find myself in times of trouble...: The conditional effect of

Twisteddoodles for New Scientist



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international organisations on policy convergence".

Feedback has one final thought: we wonder if this study reflects the influence of the Ig Nobel prizes, the awards granted for research that first makes you laugh and then makes you think. The Igs have been running since 1991, which is more than enough time for scientists and funders to realise that they are a potential avenue to scientific success. This study, surely, will be in the running for an Ig at some point – but did the researchers foresee that?

We all know what an Oscar bait movie is: it's probably a historical drama, its lead actor undergoes some sort of difficult physical transformation and it's not in the least bit funny. Is this study of Beatles lyrics in scientific papers perhaps an example of Ig Nobel bait?

Love me do

Speaking of the Ig Nobels, this year's prizes were bestowed on 3 September. The winners came together at a ceremony in Zurich, Switzerland. It was the first time the Ig Nobel ceremony had been held outside the US, on the grounds that "it has become unsafe for our guests to visit the country", said founder Marc Abrahams. Eesh.

Ten prizes were awarded in total. The biomechanics prize went to a three-person team that developed a new definition of kissing. Kissing, it seems, is "non-agonistic interactions involving directed, intraspecific, oral-oral contact with some movement of the lips/mouthparts and no food transfer". Feedback's immediate thought was that this seemed to exclude kissing people on the cheek, and indeed it does. Then

we delved into the paper and discovered there is a possibility that kissing evolved from premastication: chewing food in your mouth before sharing it with someone else. We should have known better than to read on.

Can't buy me love

We will pass swiftly over the economics prize, which recognises work showing that upper-class people are more likely to engage in unethical behaviour. Thanks, we know; the more pressing issue is stopping them.

Instead, we will move to the biology prize, given to the authors of a 2024 study "for gently stepping on various parts of various venomous snakes – repeatedly, at various temperatures at various times of day – to try to learn what does and what does not induce a venomous snake to bite a human".

Feedback actually found ourselves having a sense-of-humour failure about this: we get the idea, but isn't it a bit, well, cruel? So we looked it up and learned that the editors retracted the study in February 2025, on the grounds that "The Animal Ethics Committee of Butantan Institute confirmed that the approval granted to the Authors for their research on snakes (*Bothrops jararaca*) did not include newborn snakes or the use of the 'soft stepping' method." Maybe Feedback is soft-hearted, but we find ourselves agreeing that it is bad to step on baby snakes.

That was unexpectedly nasty, so let's end with the peace prize, which is "for documenting that we value friends who are vicious to people we dislike". Researchers conducted surveys showing that people prefer friends who are nice to them and who are generally nice to other people, but they also sometimes value a friend who is prepared to be mean to one's enemies. Feedback congratulates the researchers for figuring out something that one could learn by watching any romantic comedy. Or, to quote Willow Rosenberg: "That's the best friend's job: vilifying and grousing". ■

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